

A REALIZATION OF G_2 FOR LITT PROBLEM NO. 13

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ABSTRACT. This paper constructs a nonempty family of complex abelian threefolds P with integral symmetric ample divisors $X \subset P$ whose full convolution Tannaka group is the simple group of type G_2 . The construction uses cyclic triple covers of elliptic curves

$$C : w^3 = y - c, \quad c \neq 0,$$

with $y^2 = x^3 + Ax + B$. The induced Prym polarization has type $(1, 1, 3)$. Each theta section is normal and has one simple elliptic singularity of type $\tilde{E}_7$, and $\chi(\mathrm{IC}_X) = 7$. On a nonempty Zariski-open part of the parameter space, for $K = \mathrm{IC}_X$ one moreover has

$$K \text{ is a direct summand of } \mathrm{Alt}^2(K)$$

in the convolution Tannakian category. This summand comes from two components of a generic pair-incidence curve. Their separation is governed by an exact norm-conic identity over a rational function field: the double cover ordering the two residual lines is the constant quadratic cover attached to $\mathbf{Q}(\zeta_3)/\mathbf{Q}$. A generic unitary twist gives Hodge multiplicities $(2, 3, 2)$. Together with the exterior-square summand, these weights identify the full Tannaka group with G_2 on its standard seven-dimensional representation. This gives an affirmative answer to the G_2 case of Litt Problem No. 13.

1. INTRODUCTION

Let A be a complex abelian variety. Convolution of constructible complexes on A , after quotienting by negligible objects, gives a neutral Tannakian category. An integral subvariety $Z \subset A$ with nonzero intersection-cohomology Euler characteristic therefore determines an algebraic group $G(\mathrm{IC}_Z)$ and a faithful representation of dimension $\chi(\mathrm{IC}_Z)$; see [15]. Litt Problem No. 13 asks whether each of the exceptional simple types

$$G_2, \quad E_7, \quad E_8, \quad F_4$$

occurs in this way [17]. The result below settles the G_2 case. No claim is made here for the other three types. It is worth stressing that the singular theta surface is essential to the construction: the exclusion results for smooth subvarieties in [14] do not apply to this case.

The construction is defined over a rational parameter space. Put

$$\Delta_E = 4A^3 + 27B^2, \quad \Delta_{\mathrm{br}} = 4A^3 + 27(B - c^2)^2,$$

and let

$$\mathcal{B}_{\mathbf{Q}} = \mathrm{Spec} \mathbf{Q}[A, B, c, c^{-1}, \Delta_E^{-1}, \Delta_{\mathrm{br}}^{-1}], \quad \mathcal{B} = \mathcal{B}_{\mathbf{Q}} \times_{\mathbf{Q}} \mathbf{C}.$$

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For $b = (A, B, c) \in \mathcal{B}(\mathbf{C})$, let

$$E_b : y^2 = x^3 + Ax + B, \quad C_b : w^3 = y - c.$$

Here C_b denotes the smooth projective cyclic cover defined by the section $y - c \in H^0(E_b, \mathcal{O}_{E_b}(3O))$; the displayed equation is its affine chart. The reduced branch divisor makes this cover connected. Let

$$\pi_b : C_b \longrightarrow E_b, \quad (x, y, w) \longmapsto (x, y)$$

be the cyclic triple cover. The condition $\Delta_{\text{br}} \neq 0$ means that the line $y = c$ meets E_b in three distinct points; these are precisely the branch points of π_b . Write $O \in E_b$ for the point at infinity and set

$$\kappa_b = \pi_b^* \mathcal{O}_{E_b}(O), \quad P_b = \ker(\text{Nm}_{\pi_b})^0.$$

The centred theta divisor on $J(C_b)$ restricts to the divisor

$$X_b = \{M \in P_b : h^0(C_b, \kappa_b \otimes M) > 0\}.$$

Write $\mathcal{P} \rightarrow \mathcal{B}$ and $\mathcal{X} \subset \mathcal{P}$ for the resulting relative Prym and relative theta section.

Theorem 1.1 (Main theorem). *There is a nonempty Zariski-open subset $\mathcal{B}^\circ \subset \mathcal{B}$ such that for every $b \in \mathcal{B}^\circ(\mathbf{C})$ the following hold.*

- (i) P_b is an abelian threefold, and the polarization induced by the centred theta divisor has type $(1, 1, 3)$.
- (ii) $X_b \subset P_b$ is an integral normal symmetric ample Cartier divisor with a unique singular point at the origin; this singularity is simple elliptic of type $\tilde{E}_7$.
- (iii) $\chi(\text{IC}_{X_b}) = 7$, and there is an isomorphism of pairs

$$(G(\text{IC}_{X_b}), \omega(\text{IC}_{X_b})) \simeq (G_2, V_7),$$

as a complex reductive algebraic group together with its faithful representation. Here V_7 denotes the standard seven-dimensional representation. In particular, the full convolution Tannaka group is connected and equal to G_2 .

Moreover, $\mathcal{X} \subset \mathcal{P}$ is a symmetric relatively ample effective Cartier divisor, flat over $\mathcal{B}$, with geometrically integral fibres.

Revision note. The first release used the special equation $w^6 = x^3 + Ax + B$, corresponding to $c = 0$. That specialization carries the additional involution $w \mapsto -w$, leaving an unaddressed compatibility issue in the surface argument. The present version excludes that specialization, works with $c \neq 0$, and rederives the norm-conic and finite-field certificates for the shifted family. The former isotrivial-CM assertion for the ambient Prym family is not used and is not retained.

Proof outline. Fix $b \in \mathcal{B}(\mathbf{C})$ and write $(P, X) = (P_b, X_b)$. The norm fibre in $C^{(3)}$ resolves X . It has a single $\tilde{E}_6$ singularity; after blowing up and contracting the trigonal pencil one obtains the minimal resolution $\rho : S \rightarrow X$ with one exceptional elliptic curve of self-intersection -2 . Adjunction, the simple-elliptic geometric genus, and Noether's formula give

$$K_S^2 = 16, \quad \chi(\mathcal{O}_S) = 2, \quad e(S) = 8.$$

The semismall decomposition theorem then yields

$$\chi(\text{IC}_X) = 7, \quad \text{Stab}_P(X) = \{0\}.$$

After a generic unitary twist, the same resolution gives Hodge multiplicities $(2, 3, 2)$.

For a geometric generic point x of the relative smooth theta locus $\mathcal{M} = (\mathcal{X}/\mathcal{B})_{\text{sm}}$, let D_x be its unique effective divisor of degree three. A section of $H^0(C, 3\kappa - D_x)$ has cyclic norm equal to the fixed norm line times a conic. The determinant of that conic factors as one structural line and an integral residual octic. On the residual octic an exact identity identifies the factor-ordering double cover with the constant cover $\mathbf{Q}(\sqrt{-3})/\mathbf{Q}$. Over $\mathbf{C}$ it therefore splits into two branches. A scheme-theoretic Abel inverse turns these branches into the two nonexceptional components of the generic pair-incidence curve

$$C_x = \{y \in X : -x - y \in X\}.$$

A boundary calculation excludes further nonexceptional components.

The two generic components spread over a nonempty open $\mathcal{M}^\circ \subset \mathcal{M}$. For every fibre over the open image $\mathcal{B}^\circ \subset \mathcal{B}$, top cohomology of the pair fibres and the decomposition theorem give

$$\text{IC}_X \mid \text{Alt}^2(\text{IC}_X)$$

in the convolution Tannakian category. If $G = G(\text{IC}_X)$ and $V = \omega(\text{IC}_X)$, the remaining inputs are

$$\begin{aligned} G &\subset O(V), & \dim V &= 7, & V|_{G^0} &\text{irreducible,} \\ \mu : \mathbf{G}_m &\longrightarrow [G^0, G^0] & \text{with weights } &-2, 0, 2 \text{ of multiplicities } &2, 3, 2, \\ & & \text{Hom}_G(V, \Lambda^2 V) &\neq 0. \end{aligned}$$

These data exclude SO_7 and the seven-dimensional PGL_2 module and force $G^0 = G_2$. The same exterior-square morphism excludes the only possible extra component, hence $G = G_2$.

Exact calculations. Exact polynomial calculations are confined to the pair-incidence analysis. The supplementary certificate reconstructs the norm conic from the shifted curve, verifies the determinant factorization and the identity governing the ordering cover, and checks two characteristic-13 witnesses. The geometric arguments pass these explicit calculations to the generic characteristic-zero statements; the decomposition-theorem and group-theoretic steps do not depend on the certificate.

Organization. The proof is arranged so that each input can be checked where it is first used.

- (1) Proposition 3.2, Lemmas 3.3 and 3.5, and Corollary 3.4 establish the geometry of the theta section and $\chi(\text{IC}_X) = 7$.
- (2) Proposition 3.6 computes the generic-twist Hodge multiplicities $(2, 3, 2)$.
- (3) Propositions 4.2, 4.5 and 4.9 and Lemmas 4.3 and 4.12 identify the two generic pair components, and Theorem 5.1 spreads them.
- (4) Lemmas 6.1 to 6.3 extract an unshifted IC_X summand in alternating pair convolution.
- (5) Proposition 7.1 and Corollary 7.2 identify the connected and full Tannaka groups.

Unless a rational model or a finite-field reduction is explicitly mentioned, the geometric construction is over $\mathbf{C}$. Intersection complexes, cohomology, and Hodge modules retain their canonical $\mathbf{Q}$ -structures. The Tannakian category is obtained after extension of the underlying perverse sheaves from $\mathbf{Q}$ to $\mathbf{C}$. Thus all Tannaka groups and tautological modules below are over $\mathbf{C}$. The convention $\mathbf{P}(\mathcal{E}) = \text{Proj}(\text{Sym } \mathcal{E}^\vee)$

is used, so $\mathbf{P}(\mathcal{E})$ parametrizes lines in the fibres of $\mathcal{E}$. For an integral scheme Y , $k(Y)$ denotes its function field and $k(y)$ the residue field of a point y .

2. TANNAKIAN CONVENTIONS

Let A be a complex abelian variety and let $m : A \times A \rightarrow A$ be addition. For constructible complexes with coefficients in $\mathbf{Q}$ or $\mathbf{C}$, set

$$K * L = Rm_*(K \boxtimes L).$$

Set

$$\mathbf{S}(A) = \{Q \in \text{Perv}(A, \mathbf{C}) : \chi(A, Q) = 0\},$$

$$\mathbf{T}(A) = \{M \in D_c^b(A, \mathbf{C}) : {}^pH^i(M) \in \mathbf{S}(A) \text{ for every } i\}.$$

The first is a Serre subcategory and the second is the corresponding triangulated tensor ideal. The quotient

$$\overline{D}(A) = D_c^b(A, \mathbf{C})/\mathbf{T}(A)$$

inherits the perverse t -structure; its heart is

$$\overline{\text{Perv}}(A) = \text{Perv}(A, \mathbf{C})/\mathbf{S}(A).$$

Convolution is t -exact on $\overline{D}(A)$, and the displayed heart is a rigid symmetric monoidal neutral Tannakian category [11, Proposition 3.1]. For the pure polarizable objects used below, the tensor subcategory generated by one object is semisimple by the decomposition theorem; its Tannaka group is therefore reductive. Let δ_0 denote the skyscraper perverse sheaf at the origin; its image is the tensor unit for convolution.

For an integral subvariety $Z \subset A$ of dimension d , let

$$\text{IC}_Z = j_{!*} \mathbf{Q}_{Z_{\text{reg}}}[d], \quad j : Z_{\text{reg}} \hookrightarrow A,$$

be its perverse intersection complex, extended by zero to A . If $\chi(\text{IC}_Z) > 0$, let $G(\text{IC}_Z)$ denote the Tannaka group of the category generated by $\text{IC}_Z \otimes_{\mathbf{Q}} \mathbf{C}$ in the displayed quotient, and by $\omega(\text{IC}_Z)$ its tautological module. Then

$$\dim \omega(\text{IC}_Z) = \chi(\text{IC}_Z).$$

The action on the tautological module is faithful by construction. Scalar extension is suppressed inside Tannakian expressions. Let IC_Z^H denote the polarizable pure Hodge module whose underlying perverse $\mathbf{Q}$ -sheaf is IC_Z .

The commutativity constraint contains the usual cohomological sign. Since the support in this paper has even dimension two, the alternating projector on $\text{IC}_X * \text{IC}_X$ is carried by the fibre functor to the ordinary exterior square of $\omega(\text{IC}_X)$. Let $\text{Alt}^2(\text{IC}_X)$ denote its image in the Tannakian quotient.

3. THE SHIFTED CYCLIC THETA SECTION

Until Section 5, fix $b = (A, B, c) \in \mathcal{B}(\mathbf{C})$ and suppress it from notation. Thus

$$E : y^2 = f_3(x), \quad C : w^3 = y - c,$$

where $f_3(x) = x^3 + Ax + B$. Set

$$Y = y - c,$$

so that $w^3 = Y$. The map $\pi : C \rightarrow E$ is a cyclic cover of degree three. Its branch divisor is the reduced intersection of E with the line $y = c$.

3.1. The cover, the Prym, and the polarization. Let $b_1, b_2, b_3 \in E$ be the branch points and let $r_i \in C$ be the ramification point above b_i . Since

$$\operatorname{div}_E(y - c) = b_1 + b_2 + b_3 - 3O,$$

the branch divisor belongs to $|3O|$ and, in the group law of E ,

$$(3.1) \quad b_1 + b_2 + b_3 = O.$$

Put $\eta = \mathcal{O}_E(O)$. The cyclic covering algebra is

$$\pi_* \mathcal{O}_C = \mathcal{O}_E \oplus \eta^{-1} \oplus \eta^{-2}.$$

Riemann–Hurwitz gives $g(C) = 4$, and the canonical bundle formula gives

$$K_C = \pi^*(K_E \otimes \eta^2) = \pi^* \eta^2 = \kappa^2, \quad \kappa = \pi^* \eta.$$

By projection formula,

$$h^0(C, \kappa) = h^0(E, \eta) + h^0(E, \mathcal{O}_E) + h^0(E, \eta^{-1}) = 2.$$

The pullback $\pi^* : J(E) \rightarrow J(C)$ is injective. Indeed, if $\alpha \in \operatorname{Pic}^0(E)$ and $\pi^* \alpha \simeq \mathcal{O}_C$, then

$$1 = h^0(C, \pi^* \alpha) = h^0(E, \alpha) + h^0(E, \alpha \eta^{-1}) + h^0(E, \alpha \eta^{-2}),$$

and the last two terms vanish; hence $\alpha = \mathcal{O}_E$. By duality, $\ker(\operatorname{Nm}_\pi)$ is connected. Thus set

$$P = \ker(\operatorname{Nm}_\pi).$$

It has dimension three. If $\tau(w) = \zeta_3 w$, then $\pi^* \operatorname{Nm}_\pi = 1 + \tau + \tau^2$ on $J(C)$. Consequently $\operatorname{Im}(1 - \tau) \subset P$. The τ -invariant part of $H^0(C, \Omega_C^1)$ is $\pi^* H^0(E, \Omega_E^1)$ and has dimension one, so $\operatorname{Im}(1 - \tau)$ has dimension three. Hence

$$(3.2) \quad P = \operatorname{Im}(1 - \tau) = \ker(\operatorname{Nm}_\pi).$$

The same argument works in the relative family. Indeed $\operatorname{Nm}_\pi \circ \pi^* = [3]$ on the relative elliptic Jacobian, so the norm map is surjective; its kernel is smooth, and the fibrewise injectivity of π^* proved above makes every kernel fibre connected. Thus $\mathcal{P} = \ker(\operatorname{Nm}_\pi) \rightarrow \mathcal{B}$ is an abelian scheme of relative dimension three.

Use the theta characteristic κ to centre the theta divisor on $J(C) = \operatorname{Pic}^0(C)$:

$$\Theta_\kappa = \{M \in J(C) : h^0(C, \kappa \otimes M) > 0\}.$$

Let $L = \mathcal{O}_{J(C)}(\Theta_\kappa)|_P$. The type of the restricted polarization can be read off directly. Under the principal polarizations of the two Jacobians, π^* and Nm_π are dual homomorphisms; hence the orthogonal complement of P in $J(C)$ is $\pi^* J(E)$ [2, Section 12.3]. Therefore

$$\ker(\varphi_L) = P \cap \pi^* J(E).$$

Since π^* is injective and $\operatorname{Nm}_\pi \circ \pi^* = [3]$ on $J(E)$,

$$P \cap \pi^* J(E) = \pi^* J(E)[3].$$

This group has order 9. If the polarization type is (d_1, d_2, d_3) with $d_1 \mid d_2 \mid d_3$, then $|\ker(\varphi_L)| = (d_1 d_2 d_3)^2$; hence $d_1 d_2 d_3 = 3$ and necessarily

$$(3.3) \quad \operatorname{type}(L) = (1, 1, 3), \quad L^3 = 3! d_1 d_2 d_3 = 18.$$

Scheme-theoretically,

$$X = \Theta_\kappa \cap P = \{M \in P : h^0(C, \kappa \otimes M) > 0\}.$$

The theta divisor is preserved as a closed subscheme by the Serre-duality involution $L' \mapsto K_C \otimes (L')^{-1}$ on $\text{Pic}^3(C)$. Since $\kappa^2 = K_C$, this involution is $M \mapsto M^{-1}$ in centred coordinates. Relative Serre duality preserves the determinantal ideal of theta, so after restriction to P one obtains

$$(3.4) \quad X = -X$$

as closed subschemes. The centre will be seen to be unique in (3.8).

3.2. The norm fibre and the resolution. Let

$$n : C^{(3)} \longrightarrow \text{Pic}^3(E), \quad D \longmapsto \mathcal{O}_E(\pi_* D),$$

and put

$$N = n^{-1}(\eta^3), \quad a : N \longrightarrow X, \quad D \longmapsto \mathcal{O}_C(D) \otimes \kappa^{-1}.$$

For the centred Abel map $D \mapsto \mathcal{O}_C(D) \otimes \kappa^{-1}$ one has scheme-theoretically

$$N = C^{(3)} \times_{J(C)} P.$$

Thus a is its base change to P . The map a is surjective: if $M \in X$, a nonzero section of $\kappa \otimes M$ has an effective divisor D of degree three, and

$$\text{Nm}_\pi \mathcal{O}_C(D) = \text{Nm}_\pi(\kappa \otimes M) = \eta^3.$$

The finite morphism $C^{(3)} \rightarrow E^{(3)}$ induced by π restricts to a finite surjection

$$N \longrightarrow |\eta^3| \simeq \mathbf{P}^2.$$

Hence $\dim N = 2$, so the restriction of the centred theta section to P is nonzero. Therefore $X \in |L|$ is an effective Cartier divisor, and it is ample.

Let $\Gamma = |\kappa| \simeq \mathbf{P}^1$ and put

$$D_0 = r_1 + r_2 + r_3 \in \Gamma.$$

Lemma 3.1 (The norm fibre). *The surface N has the unique singular point D_0 , and the germ (N, D_0) is a simple elliptic singularity of type $\tilde{E}_6$. If $\beta : \hat{N} \rightarrow N$ is the blowup of D_0 , then $\hat{N}$ is smooth, its exceptional curve E_0 is elliptic with $E_0^2 = -3$, and the strict transform $\tilde{\Gamma}$ is a (-1) -curve meeting E_0 transversely in one point. Contracting $\tilde{\Gamma}$ gives a smooth surface S and a proper morphism*

$$\rho : S \longrightarrow X.$$

The image E_{exc} of E_0 satisfies $E_{\text{exc}}^2 = -2$ and maps to the origin of X .

Proof. Because the three branch points are distinct, a neighbourhood of D_0 in $C^{(3)}$ is the product of neighbourhoods of the r_i . Choose additive analytic coordinates t_i on E at b_i so that the Abel map $E^{(3)} \rightarrow \text{Pic}^3(E)$ is, after translation, given by $t_1 + t_2 + t_3$. After rescaling local parameters z_i at r_i , the cover has $t_i = z_i^3$. Hence the norm fibre has the analytic equation

$$z_1^3 + z_2^3 + z_3^3 = 0.$$

Thus (N, D_0) is the affine cone over a smooth plane cubic, hence a simple elliptic singularity of type $\tilde{E}_6$.

It remains to exclude other critical points of the norm map on N . An unramified point contributes a nonzero differential. At a ramification point the local map is $z \mapsto z^3$. For two points colliding there,

$$z_1^3 + z_2^3 = e_1^3 - 3e_1e_2$$

has no linear term, whereas for three colliding points

$$z_1^3 + z_2^3 + z_3^3 = e_1^3 - 3e_1e_2 + 3e_3$$

has a nonzero linear term. Consequently, besides D_0 , the only possible critical divisors in the norm fibre have the form $2r_i + r_j$ with $i \neq j$. Such a divisor would require

$$2b_i + b_j \sim 3O.$$

If k is the remaining index, then (3.1) gives $b_i + b_j + b_k \sim 3O$, hence $b_i = b_k$, contrary to the reducedness of the branch divisor. Thus D_0 is the unique singular point of N .

Blowing up the vertex of the cubic cone gives a smooth surface $\widehat{N}$ with exceptional elliptic curve E_0 satisfying $E_0^2 = -3$. The pencil Γ has multiplicity one at D_0 . Indeed its members are locally the divisors of $w - t$; at $t = 0$ the three branch coordinates vary linearly and nontrivially with t . Hence $\widetilde{\Gamma}$ meets E_0 transversely in one point.

Let $a_C : C^{(3)} \rightarrow J(C)$ be the Abel–Jacobi map and let Θ_C be the principal theta divisor. The canonical class formula for a symmetric product [18] gives

$$K_{C^{(3)}} \equiv a_C^* \Theta_C.$$

Since N is a fibre of $n : C^{(3)} \rightarrow \text{Pic}^3(E)$, its normal line bundle is trivial. Adjunction therefore gives

$$K_N \equiv a^* L.$$

The map a contracts Γ , so $K_N \cdot \Gamma = 0$. The cubic cone has discrepancy -1 , hence

$$K_{\widehat{N}} = \beta^* K_N - E_0.$$

Since $E_0 \cdot \widetilde{\Gamma} = 1$, one has $K_{\widehat{N}} \cdot \widetilde{\Gamma} = -1$. Adjunction on $\widetilde{\Gamma} \simeq \mathbf{P}^1$ then gives $\widetilde{\Gamma}^2 = -1$. Contracting it produces a smooth surface S and

$$E_{\text{exc}}^2 = E_0^2 + (E_0 \cdot \widetilde{\Gamma})^2 = -2.$$

The composite $\widehat{N} \rightarrow N \rightarrow X$ is constant on $\widetilde{\Gamma}$, hence factors through $\rho : S \rightarrow X$. The curve E_{exc} maps to the origin. $\square$

Proposition 3.2 (The theta surface and its resolution). *The theta section $X \subset P$ is an integral normal symmetric ample Cartier divisor with $\text{Sing}(X) = \{0\}$. The morphism $\rho : S \rightarrow X$ of Lemma 3.1 is its minimal resolution and is an isomorphism over $X \setminus \{0\}$. Its exceptional locus is the smooth elliptic curve E_{exc} with $E_{\text{exc}}^2 = -2$; in particular, $(X, 0)$ is a simple elliptic singularity of type $\widetilde{E}_7$.*

Proof. A basis of $H^0(C, K_C)$ is represented on the affine chart by

$$\frac{dx}{y}, \quad \frac{dx}{wy}, \quad \frac{dx}{w^2y}, \quad \frac{x dx}{w^2y}.$$

Indeed these forms are regular both at the ramification points and at the three points above O . After multiplication by the common factor w^2y/dx , the canonical map is

$$(x, w) \mapsto [w^2 : w : 1 : x].$$

It is birational, so C is not hyperelliptic. Among the ten quadratic monomials in these four coordinates, the relation $X_0X_2 - X_1^2 = 0$ is the only one. Indeed, after restriction to C , representatives for the remaining monomials are

$$1, x, x^2, Y; \quad w, xw, wY; \quad w^2, xw^2,$$

in the three τ -eigenspaces, and these nine functions are linearly independent over $\mathbb{C}$. Hence the unique canonical quadric is

$$X_0X_2 - X_1^2 = 0,$$

a rank-three quadric with one ruling. By geometric Riemann–Roch, every g_3^1 on a nonhyperelliptic genus-four curve determines a ruling of the canonical quadric. Hence κ is the unique g_3^1 on C . Riemann–Kempf [13] shows that the Abel map is an isomorphism over the locus $h^0 = 1$. The corresponding points of N are smooth by the critical-point calculation above, and $|\kappa|$ is the only positive-dimensional Abel fibre. It follows that $\text{Sing}(X) \subset \{0\}$.

The Riemann singularity theorem gives $\text{mult}_0(\Theta_\kappa) = h^0(C, \kappa) = 2$. The restricted theta equation is nonzero, since $\dim X = 2 < \dim P$, and lies in $\mathfrak{m}_{P,0}^2$. Thus X is singular at the origin, and

$$(3.5) \quad \text{Sing}(X) = \{0\}.$$

The Cartier divisor X is generically reduced and Cohen–Macaulay, hence reduced. It is connected because it is ample. If it were reducible, two components would meet in positive dimension, and their intersection would lie in $\text{Sing}(X)$; therefore X is integral. Being Cohen–Macaulay it satisfies S_2 , while (3.5) gives regularity in codimension one. Serre’s criterion makes X normal.

The uniqueness of the g_3^1 shows that the Abel map is an isomorphism away from Γ . Hence ρ is a resolution whose only exceptional curve is E_{exc} . Since this curve is elliptic with self-intersection -2 , the resolution is minimal and the simple-elliptic classification identifies the germ with type $\tilde{E}_7$ [16, 19]. $\square$

Lemma 3.3 (Surface invariants). *The minimal resolution S satisfies*

$$K_S^2 = 16, \quad \chi(\mathcal{O}_S) = 2, \quad e(S) = 8.$$

Proof. From $K_N \equiv a^*L$ and $X \in |L|$ one obtains

$$K_N^2 = (a^*L)^2 = L^2 \cdot [X] = L^3 = 18.$$

For the blowup of the $\tilde{E}_6$ vertex, $K_{\hat{N}} = \beta^*K_N - E_0$, so

$$K_{\hat{N}}^2 = 18 + E_0^2 = 15.$$

The contraction $\hat{N} \rightarrow S$ is the inverse of a blowup at a smooth point, and therefore $K_S^2 = 16$.

The Cartier sequence

$$0 \longrightarrow L^{-1} \longrightarrow \mathcal{O}_P \longrightarrow \mathcal{O}_X \longrightarrow 0$$

gives $\chi(\mathcal{O}_X) = 3$: on the abelian threefold P , $\chi(\mathcal{O}_P) = 0$, $\chi(L) = 3$, and $\chi(L^{-1}) = -\chi(L) = -3$. A simple elliptic singularity has local geometric genus one; equivalently, $R^1\rho_*\mathcal{O}_S$ is a skyscraper sheaf of length one [16]. Since X is normal, $\rho_*\mathcal{O}_S = \mathcal{O}_X$, and Leray gives

$$\chi(\mathcal{O}_S) = \chi(\mathcal{O}_X) - 1 = 2.$$

Noether’s formula now yields

$$e(S) = 12\chi(\mathcal{O}_S) - K_S^2 = 24 - 16 = 8.$$

$\square$

Corollary 3.4 (The intersection complex). *There is a decomposition*

$$R\rho_*\mathbf{Q}_S[2] \simeq \mathrm{IC}_X \oplus \delta_0.$$

Moreover,

$$\chi(\mathrm{IC}_X) = 7, \quad \mathrm{Stab}_P(X) = \{0\}.$$

Proof. The map $\rho : S \rightarrow X$ is semismall, with the single relevant exceptional fibre E_{exc} . Its top intersection form is the nondegenerate matrix (-2) . The semismall decomposition theorem [4, Theorem 3.4.1] gives

$$(3.6) \quad R\rho_*\mathbf{Q}_S[2] \simeq \mathrm{IC}_X \oplus \delta_0.$$

Taking Euler characteristics and using Lemma 3.3 gives

$$(3.7) \quad \chi(\mathrm{IC}_X) = e(S) - 1 = 7.$$

A translation preserving X must preserve its unique singular point, hence

$$(3.8) \quad \mathrm{Stab}_P(X) = \{0\}.$$

In particular, the origin is the unique centre of the symmetry (3.4). $\square$

Let $\Lambda_X \subset T^*P$ be the closure of the conormal bundle of X_{reg} , and let $\Lambda_0 = T_0^*P$ be the conormal fibre at the origin. Let

$$\gamma_X : \mathbf{P}(\Lambda_X) \longrightarrow \mathbf{P}(T_0^*P)$$

be the map induced by cotangent projection.

Lemma 3.5 (Conormal degree). *The characteristic cycle and conormal Gauss degree are*

$$CC(\mathrm{IC}_X) = \Lambda_X + \Lambda_0, \quad \deg(\gamma_X) = 6.$$

In particular, the conormal Gauss map is dominant.

Proof. A general hyperplane section of a $\tilde{E}_7$ singularity is of type A_3 . Its sectional Milnor number is 3, so the local Euler obstruction is -2 [16, 19, 3]. On the other hand, the stalk of $R\rho_*\mathbf{Q}_S[2]$ at the origin has Euler characteristic $e(E_{\mathrm{exc}}) = 0$, whereas the summand δ_0 has stalk Euler characteristic 1. Thus (3.6) gives $\chi((\mathrm{IC}_X)_0) = -1$. The Dubson–Kashiwara local index formula [12] therefore gives

$$CC(\mathrm{IC}_X) = \Lambda_X + m\Lambda_0, \quad -1 = -2 + m,$$

hence $m = 1$. The Gauss map of $\Lambda_0 = T_0^*P$ is the identity on $\mathbf{P}(T_0^*P)$ and has degree one. The Franecki–Kapranov index theorem [6, Theorem 1.3], together with (3.7), then gives

$$7 = \deg(\gamma_X) + 1.$$

Thus

$$(3.9) \quad \deg(\gamma_X) = 6.$$

$\square$

3.3. The generic-twist Hodge vector. Put $U = P \setminus X$. Since X is ample, U is affine. Artin vanishing for the perverse sheaf $\mathbf{Q}_U[3]$ [1, Théorème 4.1.1] gives $H_c^i(U, \mathbf{Q}) = 0$ for $i < 3$. The long exact sequence for (P, X) therefore identifies $H^1(P, \mathbf{Q})$ with $H^1(X, \mathbf{Q})$.

The surface X is normal, is a local complete intersection in P , and has only the isolated singular point 0. Hence $\mathbf{Q}_X[2]$ and IC_X are perverse, agree on X_{reg} , and their perverse kernel and cokernel are supported at 0. Point-supported perverse sheaves have hypercohomology only in degree zero, so

$$H^1(X, \mathbf{Q}) = \mathbb{H}^{-1}(X, \mathbf{Q}_X[2]) \xrightarrow{\sim} \mathbb{H}^{-1}(X, \mathrm{IC}_X) = IH^1(X, \mathbf{Q}).$$

Taking degree -1 hypercohomology in (3.6) gives

$$H^1(P, \mathbf{Q}) \xrightarrow{\sim} IH^1(X, \mathbf{Q}) \xrightarrow{\sim} H^1(S, \mathbf{Q}).$$

Thus $q(S) = 3$, and $\mathrm{Pic}^0(P) \rightarrow \mathrm{Pic}^0(S)$ is an isogeny. Since the image of $S \rightarrow P$ is two-dimensional, S has maximal Albanese dimension.

Let $j : S \xrightarrow{\rho} X \hookrightarrow P$. A unitary rank-one character of $\pi_1(P)$ determines, via its unitary flat connection, a point $\alpha \in \mathrm{Pic}^0(P)$; write $\mathcal{L}_\alpha$ for the corresponding local system. Using the isogeny $\mathrm{Pic}^0(P) \rightarrow \mathrm{Pic}^0(S)$ proved above, coherent generic vanishing gives a nonempty Zariski-open subset of $\mathrm{Pic}^0(P)$ on which

$$H^i(S, K_S \otimes \alpha^{\pm 1}|_S) = 0 \quad (i > 0).$$

Perverse generic vanishing gives a nonempty Zariski-open subset of the Betti character torus on which

$$H^i(P, Rj_* \mathbf{Q}_S[2] \otimes \mathcal{L}_\alpha) = 0 \quad (i \neq 0).$$

The compact unitary character torus is Zariski dense in the Betti character torus (a Laurent polynomial vanishing on the unitary torus vanishes identically), and the unitary-character identification with $\mathrm{Pic}^0(P)$ is a real-analytic isomorphism. Consequently the two generic-vanishing conditions cut out Euclidean open dense subsets of the same compact unitary torus. Let Ω be their nonempty intersection.

Proposition 3.6 (Generic-twist Hodge vector). *For every $\alpha \in \Omega$, the Hodge structure $H^0(P, \mathrm{IC}_X^H \otimes \mathcal{L}_\alpha)$ has Hodge numbers*

$$(h^{2,0}, h^{1,1}, h^{0,2}) = (2, 3, 2).$$

Equivalently, the associated antidiagonal Hodge cocharacter has weights $-2, 0, 2$ on $\omega(\mathrm{IC}_X)$ with multiplicities $2, 3, 2$, respectively.

Proof. For $\alpha \in \Omega$, coherent generic vanishing [8], Riemann–Roch, and Serre duality give

$$h^0(S, K_S \otimes \alpha|_S) = 2, \quad h^2(S, \alpha|_S) = 2.$$

The complex $Rj_* \mathbf{Q}_S[2]$ is perverse, and generic vanishing for perverse sheaves on abelian varieties [15, Theorem 1.1] gives

$$H^k(S, j^* \mathcal{L}_\alpha) = 0 \quad (k \neq 2).$$

For a unitary local system the twisted Dolbeault decomposition reads

$$H^{p,q}(S, j^* \mathcal{L}_\alpha) = H^q(S, \Omega_S^p \otimes \alpha|_S), \quad p + q = 2.$$

Its total dimension is $e(S) = 8$. Thus the Hodge vector of $Rj_* \mathbf{Q}_S^H[2]$ is $(2, 4, 2)$. The Hodge-module refinement of (3.6), extended to P , is

$$Rj_* \mathbf{Q}_S^H[2] \simeq \mathrm{IC}_X^H \oplus \mathbf{Q}_0^H(-1).$$

The point summand has type $(1, 1)$, so removing it leaves $(2, 3, 2)$ for IC_X^H . $\square$

4. THE GENERIC PAIR-INCIDENCE SPLITTING

Return to the family over $\mathcal{B}$. Let $\mathcal{E} \rightarrow \mathcal{B}$ be the Weierstrass elliptic scheme with zero section O , put $\eta = \mathcal{O}_{\mathcal{E}}(O)$, and let

$$\pi : \mathcal{C} \longrightarrow \mathcal{E}, \quad \kappa = \pi^* \eta$$

be the relative shifted cyclic cover and its theta characteristic. Let $\mathcal{P} \rightarrow \mathcal{B}$ be the relative Prym scheme and $\mathcal{X} \subset \mathcal{P}$ the relative centred theta section. For any $Y \rightarrow \mathcal{B}$, undecorated symbols E_Y, C_Y, P_Y, X_Y, η and κ denote the corresponding base changes.

The relative theta divisor determined by κ restricts to a section of a relatively ample line bundle on the abelian scheme $\mathcal{P} \rightarrow \mathcal{B}$. Its restriction to every fibre is the nonzero divisor constructed in [Proposition 3.2](#). Hence its zero scheme $\mathcal{X} \subset \mathcal{P}$ is a symmetric relative effective Cartier divisor; because $\mathcal{P} \rightarrow \mathcal{B}$ is smooth, $\mathcal{X} \rightarrow \mathcal{B}$ is flat. The same proposition shows that every geometric fibre is integral and has singular locus equal to the origin. The relative smooth locus is therefore

$$\mathcal{M} = (\mathcal{X}/\mathcal{B})_{\mathrm{sm}} = \mathcal{X} \setminus 0.$$

The morphism $\mathcal{M} \rightarrow \mathcal{B}$ is smooth and surjective with geometrically integral fibres. Since $\mathcal{B}$ is integral, $\mathcal{M}$ is integral.

Let $\mathcal{N}$ be the relative norm fibre in $\mathcal{C}^{(3)}$. Every Abel class on $\mathcal{M}$ has $h^0 = 1$, and the relative Abel map identifies $\mathcal{N} \times_{\mathcal{X}} \mathcal{M}$ with $\mathcal{M}$. The universal divisor on the relative symmetric cube therefore gives a universal effective divisor of degree three over $\mathcal{M}$.

4.1. A dominant normalized chart.

Lemma 4.1 (Dominant normalized chart). *There is a nonempty open subscheme*

$$T_{\mathbf{Q}} \subset \mathrm{Spec} \mathbf{Q}[c, c^{-1}, m, m^{-1}, e_1, e_2, e_3],$$

hence an integral five-dimensional rational scheme, such that with $\lambda = m^{-3}$, after base change to $\mathbf{C}$ suitable dense opens of

$$T = T_{\mathbf{Q}} \times_{\mathbf{Q}} \mathbf{C} \quad \text{and} \quad \mathcal{M}$$

are isomorphic. On these opens the universal divisor is described by (4.2) below. On $e_1 \neq 0$, put

$$(4.1) \quad \delta = \frac{c}{e_1^3}, \quad \Lambda = e_1^3 \lambda, \quad E_2 = \frac{e_2}{e_1^2}, \quad E_3 = \frac{e_3}{e_1^3}, \quad Z = e_1 z.$$

The residual-octic family and its factor-ordering cover are defined over $T_{\mathbf{Q}}$ and are birationally pulled back from the normalized chart $e_1 = 1$.

Proof. Let D be the unique reduced degree-three divisor representing a point of a dense open in $\mathcal{M}$. Its norm on E is a unique nonvertical line

$$y = mx + n, \quad m \neq 0.$$

Set

$$x' = mx + n - c, \quad Y' = y - c, \quad w' = w.$$

Then the norm line is $Y' = x'$. If the three selected lifts have w' -coordinates α, β, γ , let e_i be their elementary symmetric functions and write

$$p(W) = W^3 - e_1 W^2 + e_2 W - e_3.$$

Set

$$\Sigma_1 = e_1^3 - 3e_1 e_2 + 3e_3, \quad \Sigma_2 = e_2^3 - 3e_1 e_2 e_3 + 3e_3^2.$$

Since $\lambda = m^{-3}$, the curve and the selected divisor are reconstructed in the primed coordinates by

$$(4.2) \quad \begin{aligned} (Y' + c)^2 &= G(x') \\ &:= (x' + c)^2 + \lambda((x')^3 - \Sigma_1(x')^2 + \Sigma_2 x' - e_3^3), \\ D : \quad x' = Y' &= (w')^3, \quad p(w') = 0. \end{aligned}$$

Conversely these formulas recover the original short-Weierstrass model. Indeed write

$$g_2 = 1 - \lambda \Sigma_1, \quad g_1 = 2c + \lambda \Sigma_2, \quad g_0 = c^2 - \lambda e_3^3, \quad n_0 = -\frac{g_2}{3\lambda}.$$

Since $m^3 \lambda = 1$, the substitution $x' = mx + n_0$ gives

$$(Y' + c)^2 = x^3 + Ax + B,$$

where

$$A = m \left(g_1 - \frac{g_2^2}{3\lambda} \right), \quad B = G(n_0).$$

The original norm line is then $y = mx + n_0 + c$. These constructions are inverse on dense opens. They are defined over $\mathbf{Q}$, and therefore construct $T_{\mathbf{Q}}$ with

$$(4.3) \quad \mathbf{C}(T) = \mathbf{C}(\mathcal{M}).$$

The equations are equivariant for

$$(x', Y', w', c) \mapsto (r^3 x', r^3 Y', r w', r^3 c),$$

which in the original coordinates is the usual Weierstrass scaling

$$(x, y, w, c, A, B) \mapsto (r^2 x, r^3 y, r w, r^3 c, r^4 A, r^6 B).$$

Thus

$$(c, \lambda, e_1, e_2, e_3, m) \mapsto (r^3 c, r^{-3} \lambda, r e_1, r^2 e_2, r^3 e_3, r m).$$

The basis of $H^0(C, 3\kappa - D)$ used below has weights 3, 3, 4, so its projective coordinates transform by

$$(4.4) \quad [u : v : z] \mapsto [u : v : r^{-1} z].$$

Taking $r = e_1^{-1}$ gives (4.1) and

$$\mathbf{Q}(c, \lambda, e_1, e_2, e_3) = \mathbf{Q}(\delta, \Lambda, E_2, E_3)(e_1).$$

If $q_0 = m/e_1$, then

$$(4.5) \quad \mathbf{Q}(T_{\mathbf{Q}}) = \mathbf{Q}(\delta, \Lambda, E_2, E_3)(q_0, e_1), \quad q_0^3 = \Lambda^{-1}.$$

This is the required normalized description. $\square$

4.2. The norm conic. All formulas in this subsection are first established over the rational function fields indicated below. Put $Y = y - c = w^3$. The evaluation map

$$H^0(C, 3\kappa) \longrightarrow H^0(D, (3\kappa)|_D)$$

has rank three and $h^0(C, 3\kappa) = 6$. Its kernel has basis

$$s_0 = x - Y, \quad s_1 = Y - e_1w^2 + e_2w - e_3, \quad s_2 = (x - e_3)w - e_1Y + e_2w^2.$$

For $s = us_0 + vs_1 + zs_2$, write

$$s = \alpha + \beta w + \gamma w^2,$$

where

$$\alpha = u(x - Y) + v(Y - e_3) - e_1zY,$$

$$\beta = e_2v + z(x - e_3),$$

$$\gamma = -e_1v + e_2z.$$

Since $w^3 = Y$, the cyclic norm is

$$(4.6) \quad \text{Nm}(s) = \alpha^3 + \beta^3Y + \gamma^3Y^2 - 3\alpha\beta\gamma Y.$$

It is divisible on E by the fixed norm line $Y - x$. The quotient is a section of $\mathcal{O}_E(6O)$ and hence has a unique expression

$$Q_s = q_{00} + q_{01}x + q_{02}Y + q_{11}x^2 + q_{12}xY + q_{22}Y^2.$$

Let

$$M_s = \begin{pmatrix} q_{00} & q_{01}/2 & q_{02}/2 \\ q_{01}/2 & q_{11} & q_{12}/2 \\ q_{02}/2 & q_{12}/2 & q_{22} \end{pmatrix}.$$

Proposition 4.2 (Determinant factorization). *Put*

$$K_0 = \mathbf{Q}(c, \lambda, e_1, e_2, e_3).$$

In $K_0[u, v, z]$ one has

$$(4.7) \quad \det(M_s) = (u - v)F_8,$$

where F_8 is homogeneous of degree eight and $u - v \nmid F_8$.

Proof. The assertion may be checked on the dense locus $e_1 \neq 0$. Apply the scaling of [Lemma 4.1](#) with $r = e_1^{-1}$; the section coordinates transform as in (4.4), so the determinant factorization pulls back from the normalized chart $e_1 = 1$. On that chart, substitute (4.6), reduce by the shifted elliptic equation, and express the quotient in the basis

$$1, x, Y, x^2, xY, Y^2$$

of $H^0(E, 6O)$. The exact calculation reconstructed in [Appendix A](#) gives a homogeneous determinant of degree nine, with $u - v$ as a factor of multiplicity one and a homogeneous degree-eight quotient. Pulling the identity back along the scaling proves it in the function field K_0 . $\square$

Lemma 4.3 (Geometric integrality of the residual octic). *The projective curve*

$$\mathcal{R} = V(F_8) \subset \mathbf{P}_{K_0}^2$$

is geometrically integral.

Proof. Reduce the normalized specialization

$$(c, \lambda, e_1, e_2, e_3) = (1, 1, 1, 0, 2)$$

modulo 13. The corresponding short-Weierstrass curve is

$$y^2 = x^3 + 2x + 5,$$

whose discriminant is nonzero; the branch cubic obtained from $y = 1$ is also separable. Multiplying F_8 by the nonzero scalar 4, set

$$H(u, v) = 4F_8(u, v, 1)$$

at this specialization. It has total degree eight and v -degree eight, with constant leading coefficient 7. At $u = 4$ its monic slice is

$$\begin{aligned} h(v) &= v^8 - 3v^7 - 2v^6 + 3v^5 + 4v^4 + 4v^3 \\ &\quad - 3v^2 - 5v - 2 \in \mathbf{F}_{13}[v]. \end{aligned}$$

Exact arithmetic gives

$$\gcd(h, v^{13^4} - v) = 1, \quad v^{13^8} - v \equiv 0 \pmod{h}.$$

Rabin's criterion therefore makes h irreducible over $\mathbf{F}_{13}$. If H factored in $\mathbf{F}_{13}[u, v]$, the constant leading coefficient in v would force the leading v -coefficients of the factors to be units. A positive- v -degree factor would retain its degree after $u = 4$, while a v -degree-zero factor would itself be a unit. Both possibilities contradict the irreducibility of h . Hence H is irreducible.

The affine curve $H = 0$ contains the rational point $(u, v) = (1, 6)$, where

$$(\partial H / \partial u, \partial H / \partial v) = (2, 0).$$

Thus the point is smooth. If the irreducible projective octic were not geometrically irreducible, Frobenius would permute its geometric components transitively, and a rational point would lie on all of them; it would then be singular. The special fibre is therefore geometrically integral.

Spread the coefficients over a localization $R_0 = \mathbf{Z}[1/N]$ with $13 \nmid N$, and choose an affine open

$$\mathcal{U} \subset \mathbb{A}_{R_0}^5 = \operatorname{Spec} R_0[c, \lambda, e_1, e_2, e_3]$$

containing the displayed characteristic-13 point on which the required denominators and one coefficient of F_8 are units. After multiplying by a unit, F_8 defines a flat projective relative Cartier divisor of degree eight in $\mathbf{P}_{\mathcal{U}}^2$. Geometric integrality is open in such a family [9, IV₃, Théorème 12.2.4(viii)]. Since the displayed special fibre is geometrically integral, so is the generic fibre $\mathcal{R}$. $\square$

Lemma 4.4 (Normalized and full function fields). *Work over $e_1 \neq 0$ and put*

$$K_n = \mathbf{Q}(\delta, \Lambda, E_2, E_3).$$

On the affine section chart $v \neq 0$, put

$$U = u/v, \quad z_0 = z/v, \quad \zeta = e_1 z_0.$$

Let $\mathcal{R}_n$ be obtained by setting $e_1 = v = 1$ and using coordinates $(\delta, \Lambda, E_2, E_3; U, \zeta)$. Under the parameter map of Lemma 4.1, let

$$\mathcal{R}_{T_{\mathbf{Q}}} = \mathcal{R} \times_{\operatorname{Spec} K_0} \operatorname{Spec} \mathbf{Q}(T_{\mathbf{Q}}).$$

Then

$$(4.8) \quad k(\mathcal{R}) = k(\mathcal{R}_n)(e_1), \quad k(\mathcal{R}_{T_{\mathbf{Q}}}) = k(\mathcal{R}_n)(e_1, q_0), \quad q_0^3 = \Lambda^{-1}.$$

The curve $\mathcal{R}_n$ is geometrically integral. After base change to $\mathbf{C}$, the factor-ordering cover on the full chart is the pullback of the normalized one through these field extensions.

Proof. The scaling in Lemma 4.1, together with (4.4), identifies the locus $e_1 v \neq 0$ birationally with $\mathbf{G}_m \times \mathcal{R}_n$. This gives the first equality in (4.8). The chart $v \neq 0$ is dense because $\mathcal{R}$ is a geometrically integral octic. Geometric integrality of $\mathcal{R}_n$ follows by faithfully flat descent from Lemma 4.3. The second equality is (4.5). Formation of the two line factors of a rank-two conic commutes with field extension, which proves the last assertion. $\square$

Proposition 4.5 (Cyclotomic ordering cover). *On the normalized chart $e_1 = v = 1$, rename*

$$(\delta, \Lambda, E_2, E_3, U, \zeta) \quad \text{as} \quad (c, \lambda, e_2, e_3, u, z)$$

and put

$$K_1 = \mathbf{Q}(c, \lambda, e_2, e_3, u).$$

Let

$$\tilde{F}(z) = F_8(u, 1, z), \quad \ell = \text{lc}_z(\tilde{F}), \quad F = \ell^{-1} \tilde{F}.$$

Then F is monic of degree six. If

$$\Delta = -4(q_{00}q_{11} - (q_{01}/2)^2),$$

there are polynomials $A_c, B_c \in K_1[z]$ with $\deg A_c = 3$ and $\deg B_c \leq 2$ such that

$$(4.9) \quad F = A_c^2 + 3\Delta B_c^2.$$

If

$$L = K_1[z]/(F) = k(\mathcal{R}_n),$$

then

$$(4.10) \quad [\Delta] = [-3] \quad \text{in } L^\times/L^{\times 2}, \quad L(\sqrt{\Delta}) = L(\sqrt{-3}).$$

There is a dense normal open $\mathcal{R}_n^\circ$ in the rank-two locus for which the intrinsic factor-ordering cover is

$$(4.11) \quad \text{Ord}_{\text{fac}, \mathcal{R}_n^\circ} \simeq \mathcal{R}_n^\circ \times_{\mathbf{Q}} \text{Spec } \mathbf{Q}(\sqrt{-3}).$$

The same statement holds after pullback to a dense normal open of $\mathcal{R}_{T\mathbf{Q}}$. In particular, the ordering cover splits over $\mathbf{C}$.

Proof. The formulas in Appendix A reconstruct F , Δ , and the coefficients of A_c and B_c from the shifted curve and verify (4.9) in $\mathbf{Q}(c, \lambda, e_2, e_3, u)[z]$. The normalized residual curve is geometrically integral by Lemma 4.4; hence F is irreducible and L is a field. The displayed formulas give $\deg A_c = 3$ and $\deg B_c = 2$ generically. Reducing (4.9) modulo F gives

$$-3\Delta = (A_c/B_c)^2 \neq 0$$

in L , proving (4.10) and showing that the generic conic has rank exactly two.

For a rank-two symmetric 3×3 matrix M , one has $\text{adj}(M) = c_0 a a^t$ for a nonzero vector a . Hence the ratio of any two nonzero principal cofactors is a square. The square class of a nonzero principal minor is therefore intrinsic and is precisely the quadratic character that orders the two line factors of the conic. After shrinking to a normal open on which Δ and B_c are units, write $\varrho^2 = -3$. The maps

$$s \mapsto \frac{A_c}{\varrho B_c}, \quad \varrho \mapsto \frac{A_c}{B_c s}$$

give inverse isomorphisms between the intrinsic algebra $\mathcal{O}[s]/(s^2 - \Delta)$ and the constant algebra $\mathcal{O}[\varrho]/(\varrho^2 + 3)$. This proves (4.11). The field extensions in (4.8) preserve the construction, and over $\mathbf{C}$ the constant quadratic cover is split. $\square$

Remark 4.6 (Arithmetic fibres). The quadratic field $\mathbf{Q}(\sqrt{-3}) = \mathbf{Q}(\zeta_3)$ explains the term “cyclotomic.” After spreading the rank-two open over a suitable $\mathbf{Z}[1/N]$ with $6 \mid N$, the two line factors at an $\mathbf{F}_q$ -point are individually defined over $\mathbf{F}_q$ precisely when -3 is a square in $\mathbf{F}_q$, equivalently when $q \equiv 1 \pmod{3}$. If $q \equiv 2 \pmod{3}$, they are exchanged by Frobenius and become defined over $\mathbf{F}_{q^2}$.

4.3. The scheme-theoretic Abel inverse. Over the chart T of Lemma 4.1, let $p : C_T \rightarrow T$ be the relative curve and let D be its universal divisor. After restricting T , the sheaf $p_*\mathcal{O}_{C_T}(3\kappa - D)$ is locally free of rank three. The residual equation $F_8 = 0$ defines a relative octic

$$\mathcal{R}_T \subset \mathbf{P}_T(p_*\mathcal{O}_{C_T}(3\kappa - D)).$$

After a further shrinking of T , the section F_8 is nonzero on every projective-space fibre. Thus $\mathcal{R}_T \rightarrow T$ is a flat projective relative Cartier divisor. Its generic fibre is geometrically integral by Lemmas 4.3 and 4.4. Flatness forces every irreducible component to dominate the integral base T , so the integral generic fibre gives a single component. A nilpotent section would vanish generically and hence be $\mathcal{O}_T$ -torsion, which flatness excludes. Thus $\mathcal{R}_T$ is integral.

Let $\mathcal{R}_T^{(2)} \subset \mathcal{R}_T$ be the open subset on which the residual conic has rank exactly two. The relative Hilbert scheme of line components of the universal conic is then finite étale of degree two over $\mathcal{R}_T^{(2)}$. Choosing one component determines the other and hence orders them; denote this cover by

$$\mathrm{Ord}_{\mathrm{fac}} \longrightarrow \mathcal{R}_T^{(2)}.$$

Its points will be written $([s], \ell_1, \ell_2)$, with $Q_s = \ell_1\ell_2$ up to a unit pulled back from the base.

Definition 4.7 (Admissible residual-norm locus). Let $\mathcal{G} \subset \mathcal{R}_T^{(2)}$ be the open locus on which the following conditions hold after pullback to $\mathrm{Ord}_{\mathrm{fac}}$:

- (i) the fixed divisor D is reduced;
- (ii) the fixed norm line and the two lines $\ell_i = 0$ meet E transversely, their degree-three line sections are pairwise disjoint, and the residual line sections avoid the branch divisor of $C \rightarrow E$;
- (iii) above each residual point, s vanishes on exactly one sheet of $C \rightarrow E$, and every such zero is simple;
- (iv) the two resulting degree-three zero divisors have $h^0 = 1$, and their Abel classes lie in the relative smooth locus of $\mathcal{X}$.

These conditions define an open subset of $\mathrm{Ord}_{\mathrm{fac}}$ invariant under interchanging ℓ_1 and ℓ_2 . Its image is open because the finite étale cover is open, and invariance makes it the full inverse image of that image. This defines $\mathcal{G}$.

Lemma 4.8 (Relative ordered-zero construction). *Put*

$$\mathrm{Ord}_{\mathrm{fac}, \mathcal{G}} = \mathrm{Ord}_{\mathrm{fac}} \times_{\mathcal{R}_T^{(2)}} \mathcal{G}.$$

Let $\mathrm{Ord}_{\mathrm{div}}$ be the functor over $\mathcal{G}$ whose W -points are ordered pairs of relative effective Cartier divisors (D_1, D_2) of degree three on C_W/W such that

$$\mathrm{div}(s_W) = D_W + D_1 + D_2$$

and the unordered pair $\{\pi_* D_1, \pi_* D_2\}$ is the unordered pair of line sections of E_W defined by the two components of Q_{s_W} . This functor is represented by a finite étale double cover of $\mathcal{G}$. The assignment $Q_s = \ell_1 \ell_2 \mapsto (D_1, D_2)$ is an isomorphism of finite étale double covers

$$\mathrm{Ord}_{\mathrm{fac}, \mathcal{G}} \xrightarrow{\sim} \mathrm{Ord}_{\mathrm{div}}.$$

It commutes with arbitrary base change, including nonreduced base change. Moreover, each D_i is a relative effective Cartier divisor of degree three, and $[\mathcal{O}(D_i) \otimes \kappa^{-1}]$ lies in $\mathcal{X}_{\mathrm{sm}/\mathcal{B}}$.

Proof. For the forward construction put $T' = \mathrm{Ord}_{\mathrm{fac}, \mathcal{G}}$ and write $Q_s = \ell_1 \ell_2$ in its universal ordering. By definition of $\mathcal{G}$, none of the tangency, branch, collision, multiple-zero, or Abel speciality loci excluded in [Definition 4.7](#) occurs. For the remainder of the forward construction, every object with a subscript T is pulled back to T' ; this base change is suppressed from the notation. The line ℓ_i then cuts out a relative finite étale divisor $B_i \subset E_T$ of degree three. Moreover

$$U_i = C_T \times_{E_T} B_i \longrightarrow B_i$$

is finite étale of degree three, because B_i avoids the branch divisor.

The zero over B_i is selected scheme-theoretically as follows. Étale-locally on B_i , split $U_i = B_i \sqcup B_i \sqcup B_i$ and trivialize the restriction of 3κ , using the induced trivialization of its norm line. If $a_1, a_2, a_3 \in \mathcal{O}_{B_i}$ are the three values of s , compatibility of the finite locally free norm with base change gives

$$(4.12) \quad a_1 a_2 a_3 = \mathrm{Nm}_\pi(s)|_{B_i} = 0$$

as an equality in the structure sheaf, not merely at geometric points. The collision exclusions imply that locally two values, say a_2, a_3 , are units. Equation (4.12) then forces $a_1 = 0$, also in nilpotent directions. Consequently

$$Z_i = Z(s) \times_{E_T} B_i \subset U_i$$

is locally exactly one open-and-closed sheet of U_i/B_i . Thus $Z_i \rightarrow B_i$ is finite étale of degree one and hence is an isomorphism.

Let t be a local equation for B_i in E_T . Since B_i occurs with multiplicity one in the norm divisor and the other norm factors are disjoint from it, $\mathrm{Nm}_\pi(s) = tu$ with u a unit along B_i . Étale-locally on a neighbourhood of B_i in the complement of the branch divisor, split the cover $C_T \rightarrow E_T$. On the selected sheet the other two values of s remain units, so the same local product formula shows that the selected value is t times a unit. The zero is therefore simple. The image $D_i \subset C_T$ of Z_i is finite étale of degree three over T' ; as a finite étale multisection of the smooth relative curve C_T/T' , it is a relative effective Cartier divisor.

The pairwise disjoint divisors D, D_1, D_2 lie in $Z(s)$ and have total relative degree nine, equal to $\deg(3\kappa)$. After dividing by their Cartier equations, the residual section has degree zero and is nonvanishing on every geometric fibre. Nakayama's lemma makes it a trivialization, so there is no vertical or nilpotent residual contribution and

$$(4.13) \quad Z(s) = D + D_1 + D_2$$

as relative Cartier divisors. Conversely, an ordered divisor datum determines the unique projective section $[s]$ with zero divisor $D + D_1 + D_2$, and taking its norm recovers the ordered line sections B_i and hence the ordered factors ℓ_i . Thus these assignments are inverse already at the divisor level.

It remains to compare this divisor construction with the relative Abel incidence. Let

$$\alpha : C_T^{(3)} \longrightarrow \mathrm{Pic}^3(C_T/T)$$

be the relative Abel map and let $\mathcal{F} = \mathrm{Nm}^{-1}(\eta^3) \subset \mathrm{Pic}^3(C_T/T)$. The norm fibre is the cartesian base change

$$N_T = C_T^{(3)} \times_{\mathrm{Pic}^3(C_T/T)} \mathcal{F}.$$

Let $W_3^{(1)} \subset \mathrm{Pic}^3(C_T/T)$ be the open on which $h^0 = 1$ and the pushforward of a Poincaré bundle is locally free of rank one and commutes with base change. The standard projective-bundle description of the Abel map and cohomology and base change then identify its source with the projectivization of that rank-one bundle. Thus the map is an isomorphism and remains so after arbitrary base change. Fibrewise, this is the $h^0 = 1$ case of Riemann–Kempf [13]:

$$\alpha^{-1}(W_3^{(1)}) \xrightarrow{\sim} W_3^{(1)}.$$

Base changing this isomorphism by $\mathcal{F} \rightarrow \mathrm{Pic}^3(C_T/T)$ and translating by κ^{-1} gives

$$N_T^{(1)} \xrightarrow{\sim} X_T^{(1)}.$$

Here, by definition, $X_T^{(1)}$ is the open part of the theta section represented by line bundles with $h^0 = 1$. Restrict both sides to the inverse images of $X_T^{(1)} \cap X_{T,\mathrm{sm}/T}$, as imposed above. Thus the divisor construction descends to the relative smooth Abel incidence.

Zero schemes, fibre products, finite locally free norms, and the cartesian Abel construction all commute with arbitrary base change. Hence both composites preserve the universal zero schemes and are the identity as morphisms. The use of $C_T^{(3)} \simeq \mathrm{Hilb}^3(C_T/T)$ is only representability; no normalization of the ordering cover or of pair fibres is involved. $\square$

Let $x_T : T \rightarrow \mathcal{M}$ be the point represented by the universal divisor D , so that $x_T = [\mathcal{O}(D) \otimes \kappa^{-1}]$. Define

$$\mathcal{I}_T^\dagger = \{y \in X_T^{(1)} \cap X_{T,\mathrm{sm}/T} : -x_T - y \in X_T^{(1)} \cap X_{T,\mathrm{sm}/T}\}.$$

The two relative Abel inverses give universal divisors D_1, D_2 on $C_T \times_T \mathcal{I}_T^\dagger$. Their Abel classes satisfy

$$[\mathcal{O}(D + D_1 + D_2) \otimes \kappa^{-3}] = 0 \quad \text{in } \mathrm{Pic}^0(C_T/T).$$

Thus $\mathcal{O}(D_1 + D_2)$ and $\mathcal{O}(3\kappa - D)$ differ only by a line bundle pulled back from the base. Their divisor sections determine a unique projective relative section

$$\vartheta : \mathcal{I}_T^\dagger \longrightarrow \mathbf{P}_T(p_*\mathcal{O}_{C_T}(3\kappa - D)).$$

Taking norms shows that ϑ factors through $\mathcal{R}_T$. Set

$$\mathcal{I}_T^{\mathrm{adm}} = \vartheta^{-1}(\mathcal{G}).$$

Proposition 4.9 (Abel-incidence identification). *There is an isomorphism over $\mathcal{G}$*

$$(4.14) \quad \Phi : \mathrm{Ord}_{\mathrm{fac}, \mathcal{G}} \xrightarrow{\sim} \mathcal{I}_T^{\mathrm{adm}}, \quad ([s], \ell_1, \ell_2) \longmapsto [\mathcal{O}(D_1) \otimes \kappa^{-1}].$$

Interchanging ℓ_1 and ℓ_2 corresponds to the involution $y \mapsto -x_T - y$. The isomorphism commutes with arbitrary base change.

Proof. On $\text{Ord}_{\text{fac}, \mathcal{G}}$, put

$$y_i = [\mathcal{O}(D_i) \otimes \kappa^{-1}] \in P_T.$$

The divisor equality (4.13) gives

$$(4.15) \quad x_T + y_1 + y_2 = [\mathcal{O}(D + D_1 + D_2) \otimes \kappa^{-3}] = 0.$$

Hence $y_2 = -x_T - y_1$, and the ordered-zero construction of Lemma 4.8 defines the morphism Φ . The defining conditions of $\mathcal{G}$ show that it lands in $\mathcal{I}_T^{\text{adm}}$.

Conversely, let y be a W -point of $\mathcal{I}_T^{\text{adm}}$, write $p_W : C_T \times_T W \rightarrow W$ for the projection, put $y_1 = y$ and $y_2 = -x_T - y$, and let D_1, D_2 be their relative Abel inverse divisors. Reading (4.15) from right to left gives

$$\mathcal{O}(D_1 + D_2) \simeq \mathcal{O}(3\kappa - D) \otimes p_W^* \mathcal{N}$$

for a line bundle $\mathcal{N}$ on W . The divisor section therefore gives a unique projective section $[s]$ with $\text{div}(s) = D + D_1 + D_2$; the factor pulled back from W does not affect this projective point. Since $y_i \in P_T$,

$$\text{Nm}_\pi \mathcal{O}(D_i) = \text{Nm}_\pi(\kappa) = \eta^3.$$

Thus the effective divisor $\pi_* D_i$ on E_T is cut out by a unique projective line ℓ_i . Multiplicativity of the finite locally free norm gives, up to a unit from W ,

$$\text{Nm}_\pi(s) = \ell_D \ell_1 \ell_2, \quad Q_s = \ell_1 \ell_2,$$

where ℓ_D is the fixed norm line of D . Admissibility makes the two residual factors distinct and orders them by the ordered pair (D_1, D_2) . This constructs the inverse of (4.14).

The relative Abel inverse on the $h^0 = 1$ locus, formation of zero divisors, and finite locally free norms commute with arbitrary base change. The two composites preserve the universal Cartier divisors and hence are the identity as morphisms. Swapping D_1, D_2 gives $y_1 \leftrightarrow y_2$, which is the stated incidence involution. $\square$

Lemma 4.10 (Nonemptiness of the admissible locus). *The open subscheme $\mathcal{G}$ of Definition 4.7 is nonempty in characteristic zero. It is dense in $\mathcal{R}_T$, dominates T , and its geometric generic fibre is a nonempty dense open of the geometric generic residual curve.*

Proof. An exact characteristic-13 witness is obtained on the normalized chart at

$$(c, \lambda, e_1, e_2, e_3) = (1, 1, 1, 0, 2), \quad (u : v : z) = (3 : 5 : 1).$$

In the shifted coordinates $Y = y - c$, the elliptic equation is

$$Y^2 + 2Y = x^3 - 6x^2 + x + 5 \pmod{13},$$

equivalently, after the translation $x \mapsto x + 2$,

$$y^2 = x^3 + 2x + 5.$$

The cubic on the right has discriminant 8, and the branch cubic cut out by $Y = 0$ has discriminant 4. The selected fixed divisor is defined by

$$p(w) = w^3 - w^2 - 2,$$

whose discriminant is 1. Thus both the curve and the branch divisor are smooth and D is reduced.

At the displayed section, the residual conic has coefficient vector

$$(q_{00}, q_{01}, q_{02}, q_{11}, q_{12}, q_{22}) = (5, 1, 7, 7, 0, 2)$$

and factors as

$$Q_s = 5 + x + 7Y + 7x^2 + 2Y^2 = 5(1 + 9x + Y)(1 + 12x + 3Y).$$

The fixed norm line is $Y = x$. The discriminants of the three line sections of E are, respectively,

$$4, \quad 3, \quad 5,$$

so all three intersections are transverse. The two residual lines meet at $(12, 8)$; their intersections with $Y = x$ are $(9, 9)$ and $(6, 6)$. None of these three points lies on E , so the three degree-three line sections are pairwise disjoint. Each residual line avoids the branch divisor $Y = 0$.

The canonical-model calculation in [Proposition 3.2](#) uses only that 2 and 3 are invertible and therefore remains valid for this characteristic-13 fibre: the canonical quadric is again the rank-three quadric $X_0X_2 - X_1^2 = 0$, so $|\kappa|$ is the unique g_3^1 . A norm line arising from this pencil has no x -term, whereas each of the three lines just displayed has a nonzero x -coefficient. Their degree-three inverse images are therefore not linearly equivalent to κ and hence lie in the $h^0 = 1$ Abel locus. The local norm calculation of [Lemma 3.1](#) is valid because 2 and 3 are invertible. At a residual point the cover is étale and the residual line is transverse to E ; hence the norm of s has vanishing order one. The three nonnegative vanishing orders of s on the three sheets sum to one, so exactly one sheet contains a simple zero. This verifies all conditions in [Definition 4.7](#).

For the passage to characteristic zero, enlarge the localization $R_0 = \mathbf{Z}[1/N]$ from [Lemma 4.3](#), without inverting 13, so that the chart and all admissibility conditions spread out. Work on the affine chart $v \neq 0$. The coordinate ring A_0 of a suitable affine open in the spread of $T_{\mathbf{Q}}$ is a localization of

$$R_0[c, c^{-1}, m, m^{-1}, e_1, e_2, e_3]$$

and hence is a unique factorization domain. With $U_0 = u/v$ and $Z_0 = z/v$, clear denominators in $F_8(U_0, 1, Z_0)$ and divide by its content. The resulting primitive polynomial $f \in A_0[U_0, Z_0]$ is irreducible over $\text{Frac}(A_0)$ by [Lemmas 4.3](#) and [4.4](#); Gauss's lemma makes it irreducible in $A_0[U_0, Z_0]$. Thus

$$\mathcal{Y} = \text{Spec } A_0[U_0, Z_0]/(f)$$

is integral and R_0 -flat.

Admissibility defines an open subscheme of $\mathcal{Y}$ containing the characteristic-13 witness. It therefore contains a principal open $D(g)$ with $g \neq 0$. Since $\mathcal{Y}$ is integral, $D(g)$ contains its generic point. Its characteristic-zero fibre is nonempty and dominates the generic point of $T_{\mathbf{Q}}$. After base change to $\mathbf{C}$ it lies in $\mathcal{G}$. Hence $\mathcal{G}$ is nonempty, dense in $\mathcal{R}_T$, and dominates T ; its geometric generic fibre is a nonempty open of a geometrically integral curve and is therefore dense. $\square$

Corollary 4.11 (The two admissible branches). *After replacing $\mathcal{G}$ by a dense open subset, there is an isomorphism over $\mathcal{G}$*

$$\mathcal{I}_T^{\text{adm}} \simeq \mathcal{G} \sqcup \mathcal{G}.$$

The incidence involution exchanges the two copies. On a geometric generic fibre over T , each copy is a dense open of a geometrically integral curve.

Proof. By [Proposition 4.5](#) and [Lemma 4.4](#), the finite étale factor-ordering cover has two rational sections over $\mathbf{C}$ at the generic point of $\mathcal{G}$. On the principal-minor

chart, after fixing $\varrho \in \mathbf{C}$ with $\varrho^2 = -3$, they are given by $s = \pm A_c/(\varrho B_c)$ for the intrinsic coordinate $s^2 = \Delta$. They extend to two disjoint sections after shrinking $\mathcal{G}$, so $\text{Ord}_{\text{fac}, \mathcal{G}} \simeq \mathcal{G} \sqcup \mathcal{G}$. Apply [Proposition 4.9](#). The final assertion follows from [Lemmas 4.3 and 4.10](#). $\square$

4.4. Boundary exhaustion and global labels. Let ξ be the generic point of the integral scheme $\mathcal{M}$, and let $\bar{\xi}$ be a geometric generic point. For a geometric point x of $\mathcal{M}$, define the pair fibre scheme-theoretically by

$$C_x = X \cap (-x - X) \subset P, \quad C_x^{\text{ne}} = C_x \setminus \{0, -x\}.$$

Over $\bar{\xi}$, the unique effective divisors representing $x, y, -x - y$ satisfy

$$(4.16) \quad D_x + D_y + D_{-x-y} = \text{div}(s_y)$$

for a projective section $[s_y] \in \mathbf{P}H^0(C, 3\kappa - D_x)$. This defines a rational map

$$\Psi : C_x \dashrightarrow \mathbf{P}H^0(C, 3\kappa - D_x).$$

Lemma 4.12 (Boundary exhaustion). *The rational map Ψ has finite fibres wherever it is defined. The determinant component $u = v$ gives no nonexceptional incidence-curve component; its generic divisor data give only the Abel classes $y = 0, -x$. Every irreducible curve component of $C_{\bar{\xi}}^{\text{ne}}$ dominates the residual octic $\mathcal{R}$, meets its admissible open, and is the closure of one of the two branches in [Corollary 4.11](#). In particular, $C_{\bar{\xi}}^{\text{ne}}$ has exactly two irreducible components.*

Proof. For a fixed section, the residual zero-cycle has only finitely many degree-three subcycles, so Ψ has finite fibres. In the basis used for [\(4.7\)](#), put

$$h = x - e_1 w^2 + e_2 w - e_3.$$

Since $Y = w^3$ on C , direct substitution gives

$$(4.17) \quad s_0 + s_1 = h, \quad s_2 = wh.$$

The zero divisor of $h \in H^0(C, 2\kappa)$ is $D_x + D_{-x}$. Indeed, it contains D_x , and the residual degree-three divisor has Abel class $-x$ and is unique on the generic $h^0 = 1$ locus. Equation [\(4.17\)](#) identifies the line $u = v$ with

$$\mathbf{P}(h \cdot H^0(C, \kappa)) = \mathbf{P}H^0(C, 3\kappa - D_x - D_{-x}).$$

Its members have divisor $D_x + D_{-x} + D_\kappa$ with $D_\kappa \in |\kappa|$. For a general member of this pencil, $\pi_* D_{-x}$ and $\pi_* D_\kappa$ are reduced, disjoint line sections of the plane cubic E . A degree-three subcycle using two points from the first section and one from the second cannot itself be a line section: a line through the first two points is the first line and cannot contain the third point. The same argument applies with the roles reversed. Hence the only degree-three subcycles whose pushforward is a line section are the two complete triples. Their Abel classes are $-x$ and 0 .

The exceptional parameters of the structural pencil form a finite set. If an incidence-curve component mapped dominantly to $u = v$, its generic point would contradict the preceding paragraph; if it mapped to one parameter, it would contradict the finiteness of the fibres of Ψ . Thus $u = v$ gives no nonexceptional curve component.

The same two-line argument shows that, over $\mathcal{G}$ the only Prym degree-three subcycles of the six residual zeros are the two complete line sections. By [Proposition 4.9](#) and [Corollary 4.11](#), these give exactly the two displayed admissible branches.

By the uniqueness of the g_3^1 , the only points where an Abel inverse in the definition of Ψ is not unique are $y = 0$ and $y = -x$. Thus the indeterminacy of Ψ is contained in these two exceptional points. By (4.7), every nonexceptional curve component therefore maps to the residual octic. Its image is one-dimensional because Ψ has finite fibres, and hence is dense because $\mathcal{R}$ is geometrically integral. It must meet the dense open $\mathcal{G}$. An irreducible curve cannot contain both disjoint admissible branches as open subsets, so their closures are exactly the two nonexceptional components. Here $\mathbf{C}(T) = \mathbf{C}(\mathcal{M})$ from (4.3) identifies the two geometric generic fibres. $\square$

Proposition 4.13 (The geometric generic pair fibre). *The scheme $C_{\bar{\xi}}$ is reduced, and $C_{\bar{\xi}}^{\text{ne}}$ has exactly two geometrically integral irreducible components $A_{\bar{\xi}}$ and $B_{\bar{\xi}}$. They are exchanged by $y \mapsto -x - y$, and both descend to $k(\xi)$.*

Proof. By Lemmas 4.3, 4.10 and 4.12, Propositions 4.5 and 4.9, and Corollary 4.11, the nonexceptional geometric generic fibre has exactly two geometrically integral components, exchanged by the incidence involution. Since $\text{Stab}_P(X) = \{0\}$ and x is generic, the integral Cartier divisors X and $-x - X$ are distinct. Their local equations form a regular sequence in the smooth threefold P , so $C_{\bar{\xi}}$ is a pure local complete intersection curve. It is therefore Cohen–Macaulay and has no embedded associated points. The two dense admissible branches show that it is reduced at every generic point. Hence $C_{\bar{\xi}}$ is reduced.

It remains to descend the two labels. Let τ be the order-three deck transformation. On P one has

$$1 + \tau + \tau^2 = 0,$$

and $\tau(X) = X$. Therefore

$$y_+(x) = \tau x \in C_x, \quad s_+(x) = (x, \tau x)$$

defines a rational section of the pair-incidence family; the associated ordered pair is $(\tau x, \tau^2 x)$. This section meets the smooth locus of the projection of the pair-incidence family to the x -coordinate on a nonempty open. Otherwise the Gauss covectors at τx and $\tau^2 x$ would agree for general x . Equivariance would then put the general Gauss image in the projective fixed locus of τ , contrary to (3.9). Explicitly,

$$T_0^* P = \left\langle \frac{dx}{w(w^3 + c)} \right\rangle \oplus \left\langle \frac{dx}{w^2(w^3 + c)}, \frac{x dx}{w^2(w^3 + c)} \right\rangle,$$

and the two summands have distinct τ -eigenvalues. The projective fixed locus is a point union a line, a proper subset of $\mathbf{P}^2$.

Over $k(\xi)$, the generic value of y_+ is a smooth rational point and lies on exactly one geometric component. Galois therefore fixes the component containing y_+ and, since there are exactly two components, also fixes the other. This proves descent. $\square$

5. SPREADING THE GENERIC SPLITTING

Define the relative pair-incidence scheme

$$\mathcal{I} = \{(x, y) \in \mathcal{M} \times_{\mathcal{B}} \mathcal{X} : -x - y \in \mathcal{X}\},$$

with the scheme structure induced by the two theta equations, and let $q : \mathcal{I} \rightarrow \mathcal{M}$ be the first projection. The equations $y = 0$ and $y = -x$ define two sections of q . Let $\mathcal{I}^{\text{ne}}$ denote their complement.

Theorem 5.1 (Uniform pair splitting). *There are a nonempty open subscheme $\mathcal{M}^\circ \subset \mathcal{M}$ and closed subschemes $\mathcal{A}_1, \mathcal{A}_2 \subset \mathcal{I}|_{\mathcal{M}^\circ}$ with the following properties.*

- (i) *Each $\mathcal{A}_i \rightarrow \mathcal{M}^\circ$ is flat, and every geometric fibre is a geometrically integral curve.*
- (ii) *Every geometric fibre of $q|_{\mathcal{M}^\circ}$ is reduced, and*

$$\mathcal{I}^{\text{ne}}|_{\mathcal{M}^\circ} = (\mathcal{A}_1 \cup \mathcal{A}_2) \cap \mathcal{I}^{\text{ne}}$$

scheme-theoretically.

- (iii) *The two fibre curves are distinct at every geometric point of $\mathcal{M}^\circ$, and the involution $(x, y) \mapsto (x, -x - y)$ exchanges $\mathcal{A}_1$ and $\mathcal{A}_2$.*

The image

$$\mathcal{B}^\circ = \text{im}(\mathcal{M}^\circ \rightarrow \mathcal{B})$$

is a nonempty Zariski-open subset. For every $b \in \mathcal{B}^\circ(\mathbf{C})$, the fibre

$$U_b = \mathcal{M}^\circ \times_{\mathcal{B}} \{b\}$$

is a nonempty dense connected open subset of $(X_b)_{\text{reg}}$. For every $x \in U_b(\mathbf{C})$, the curves $(\mathcal{A}_1)_x$ and $(\mathcal{A}_2)_x$ are exactly the two nonexceptional irreducible components of C_x .

Proof. Throughout the proof the schemes may be restricted to a smaller nonempty open of $\mathcal{M}$ without changing notation; the final open is denoted $\mathcal{M}^\circ$. By [Proposition 4.13](#), the geometric generic fibre of $q : \mathcal{I} \rightarrow \mathcal{M}$ is reduced and has exactly two geometrically integral components, both defined over $k(\mathcal{M})$. Let $\mathcal{A}_1, \mathcal{A}_2 \subset \mathcal{I}$ be their closures. The involution exchanges them because it exchanges their generic fibres.

The morphism q is projective and of finite presentation. Apply generic flatness to q and to $\mathcal{A}_i \rightarrow \mathcal{M}$. Geometric reducedness of the fibres of q and geometric integrality of the fibres of each $\mathcal{A}_i$ then hold after restricting to a nonempty open of $\mathcal{M}$; this is the standard spreading of these properties in a flat family (Stacks Project, Tags [052A](#), [0578](#), and [0559](#)). This proves the flatness and fibre assertions in (i), as well as the first assertion of (ii).

Since q is projective and the $\mathcal{A}_i$ are flat, their closed immersions define two sections of the relative Hilbert scheme $\text{Hilb}_{\mathcal{I}/\mathcal{M}}$. Its diagonal is closed. The equalizer of the two sections is therefore closed and omits the generic point. Removing it makes $(\mathcal{A}_1)_x$ and $(\mathcal{A}_2)_x$ distinct for every geometric x .

Give $\mathcal{A} = \mathcal{A}_1 \cup \mathcal{A}_2$ its scheme-theoretic union structure. Let $\mathcal{J}$ be the ideal of $\mathcal{A} \cap \mathcal{I}^{\text{ne}}$ in $\mathcal{I}^{\text{ne}}$. The geometric generic fibre is reduced and consists exactly of the two generic components, so $\mathcal{J}$ vanishes there. The image of $\text{Supp}(\mathcal{J})$ in the noetherian integral scheme $\mathcal{M}$ is constructible by Chevalley's theorem and does not contain its generic point. Its closure is therefore proper. Remove this closure. Then $\mathcal{J} = 0$, and

$$\mathcal{I}^{\text{ne}} = \mathcal{A} \cap \mathcal{I}^{\text{ne}}$$

over the remaining open. The fibrewise reducedness, the distinctness and geometric integrality of the two curve fibres, and this scheme-theoretic equality prove (ii) and (iii).

Denote the remaining open by $\mathcal{M}^\circ$. The morphism $\mathcal{M}^\circ \rightarrow \mathcal{B}$ is open because it is the restriction of the smooth morphism $\mathcal{M} \rightarrow \mathcal{B}$ (Stacks Project, Tag [056G](#)). Hence its image $\mathcal{B}^\circ$ is nonempty and open. For $b \in \mathcal{B}^\circ(\mathbf{C})$, the fibre U_b is a nonempty

open of the integral surface $(X_b)_{\text{reg}}$; it is therefore dense, irreducible, and connected. The fibrewise conclusions (i)–(iii) give the final assertion. $\square$

6. PAIR CONVOLUTION DETECTS IC_X

Fix $b \in \mathcal{B}^\circ(\mathbf{C})$, put $(P, X) = (P_b, X_b)$, and let $U = U_b$ as in [Theorem 5.1](#). Let $\rho : S \rightarrow X$ be the minimal resolution and let $j : S \rightarrow P$ be its composition with the inclusion. Put

$$K = \text{IC}_X, \quad \tilde{K} = Rj_* \mathbf{Q}_S[2] = K \oplus \delta_0.$$

Consider

$$\tilde{m} : S \times S \longrightarrow P, \quad (p, q) \longmapsto j(p) + j(q),$$

and let $\sigma(p, q) = (q, p)$. A superscript $-$ denotes the anti-invariant direct summand under σ .

Lemma 6.1 (Top cohomology on the split-pair locus). *After replacing U by a dense open subset, put $U^+ = -U$. For $t = -x \in U^+$, the fibre $F_t = \tilde{m}^{-1}(t)$ has exactly four one-dimensional irreducible components in its reduction: the strict transforms of $(\mathcal{A}_1)_x$ and $(\mathcal{A}_2)_x$ and the two structural components*

$$E_{\text{exc}} \times \{s_t\}, \quad \{s_t\} \times E_{\text{exc}}.$$

Here s_t is the unique point of S above t . The involution σ exchanges the first two components and exchanges the last two. Consequently

$$\dim \mathcal{H}^{-2}(\tilde{K} * \tilde{K})_{-,t} = 2, \quad \dim \mathcal{H}^{-2}(K * K)_{-,t} = 1.$$

After a further shrinking of U , the latter spaces form the constant rank-one local system on U^+ .

Proof. Fix $t \in U^+$, write $x = -t \in U$, and let $E_{\text{exc}} = \rho^{-1}(0)$. By definition, C_x parametrizes ordered pairs whose sum is $-x = t$. Hence the fibre $F_t = \tilde{m}^{-1}(t)$ has precisely four irreducible curve components: the strict transforms of $(\mathcal{A}_1)_x, (\mathcal{A}_2)_x$ and the two structural components

$$C_{1,t} = E_{\text{exc}} \times \{s_t\}, \quad C_{2,t} = \{s_t\} \times E_{\text{exc}},$$

where s_t is the unique point above t . Indeed, away from E_{exc} the resolution is an isomorphism, while a pair with one entry on E_{exc} must have the other entry over t . Thus there are no further top-dimensional components. The involution σ exchanges $(\mathcal{A}_1)_x$ with $(\mathcal{A}_2)_x$ and $C_{1,t}$ with $C_{2,t}$.

Proper base change identifies

$$\mathcal{H}^{-2}(\tilde{K} * \tilde{K})_t = H^2(F_t, \mathbf{Q}).$$

The top cohomology of a proper complex curve is freely generated by the fundamental classes of the irreducible components of its reduction. Thus $H^2(F_t, \mathbf{Q})_-$ has the explicit basis

$$[(\mathcal{A}_1)_x] - [(\mathcal{A}_2)_x], \quad [C_{1,t}] - [C_{2,t}].$$

Here and below the same notation is used for a curve and its strict transform in F_t . Holomorphic transposition preserves complex orientation, so

$$(6.1) \quad \dim \mathcal{H}^{-2}(\tilde{K} * \tilde{K})_{-,t} = 2.$$

Expand the decomposition $\tilde{K} = K \oplus \delta_0$:

$$\tilde{K} * \tilde{K} = (K * K) \oplus (K * \delta_0) \oplus (\delta_0 * K) \oplus (\delta_0 * \delta_0).$$

At $t \neq 0$, the last term has zero stalk. The two middle terms are both canonically K , and transposition exchanges them. Since t is a smooth point of the surface X , their anti-invariant stalk in degree -2 is the line generated by $[C_{1,t}] - [C_{2,t}]$. Taking dimensions in this direct sum decomposition gives

$$(6.2) \quad \dim \mathcal{H}^{-2}(K * K)_{-,t} = 1.$$

There is no Koszul sign: K has geometric shift $[2]$, and $(-1)^{2 \cdot 2} = 1$.

After a topological stratification and a further shrinking of U^+ , proper stratified local triviality identifies the four component classes in nearby fibres. Hence the total anti-invariant local system is constant of rank two, with the two displayed component differences as a global basis. The anti-invariant cross term is its constant rank-one sub-local system generated by $[C_{1,t}] - [C_{2,t}]$. The direct-sum decomposition above identifies $\mathcal{H}^{-2}(K * K)_-$ with the quotient by this sub-local system. That quotient is constant of rank one, as asserted. No choice of a complementary component class is used. $\square$

Lemma 6.2 (Vanishing away from the theta divisor). *There is a dense open subset $W \subset P \setminus X$ such that, for every $t \in W$, the scheme $X \cap (t - X)$ is a smooth connected projective curve. The involution $p \mapsto t - p$ acts as the identity on its top cohomology. In particular,*

$$\mathcal{H}^{-2}(K * K)_-|_W = 0.$$

Proof. The sum map $m|_{X \times X} : X \times X \rightarrow P$ is surjective because two translates of the ample divisor X always meet. Moreover,

$$m((\{0\} \times X) \cup (X \times \{0\})) = X.$$

Hence, for $t \notin X$, the fibre lies entirely in $X_{\text{reg}} \times X_{\text{reg}}$. Generic smoothness for the restricted sum map therefore gives smoothness of the general fibre. Since $t \notin X$ and $\text{Stab}_P(X) = \{0\}$, the translate $t - X$ does not contain X ; its restriction to the integral Cohen–Macaulay surface X is therefore an effective ample Cartier divisor. Hartshorne’s connectedness theorem, applied on X , shows that this divisor, namely $X \cap (t - X)$, is connected [10, Corollary III.7.9]. The involution $p \mapsto t - p$ preserves its complex orientation and hence acts by $+1$ on its one-dimensional top cohomology. Therefore

$$(6.3) \quad \mathcal{H}^{-2}(K * K)_- = 0 \quad \text{on a dense open subset of } P.$$

$\square$

Lemma 6.3 (Extraction of the divisor-supported summand). *In the decomposition-theorem splitting of $(K * K)_-$, the constant rank-one local system of Lemma 6.1 is the restriction of an unshifted IC_X summand. It cannot arise from a shifted simple summand with support P , or from a summand with any other support.*

Proof. The complex $(K * K)_-$ is a direct summand of the projective direct image of a pure polarizable Hodge module. By the decomposition theorem and semisimplicity [1, 20], it is a direct sum of shifted simple intersection complexes. A summand with proper support containing the generic point of U^+ must have support X . Such a simple summand has the form $\text{IC}_X(\mathcal{V})[r]$, where $\mathcal{V}$ is an irreducible local system on a dense smooth open $X_{\mathcal{V}} \subset X_{\text{reg}}$. Its restriction to $U^+ \cap X_{\mathcal{V}}$ is $\mathcal{V}[2 + r]$, so contribution to ordinary degree -2 forces $r = 0$.

A full-support summand cannot account for the line in (6.2). Choose a general point of U^+ outside the remaining singular strata of the full-support summands, take a normal analytic disk Δ to X , and write $j_\Delta : \Delta^* \hookrightarrow \Delta$. Locally, a simple perverse intersection complex with support P has the form

$$\mathcal{L}[2] \boxtimes (j_{\Delta,*} \mathcal{W}[1])$$

for local systems $\mathcal{L}$ and $\mathcal{W}$. Writing $i_0 : \{0\} \hookrightarrow \Delta$, the defining no-quotient property of intermediate extension gives

$$\mathcal{H}^0(i_0^* j_{\Delta,*} \mathcal{W}[1]) = 0.$$

Since a perverse sheaf on a disk has ordinary stalk amplitude $[-1, 0]$, $i_0^* j_{\Delta,*} \mathcal{W}[1]$ is concentrated in degree -1 when nonzero. Thus an unshifted full-support summand can contribute there only in ordinary degree -3 . Contribution in degree -2 requires the perverse-degree-one shift $[-1]$. But on the dense smooth open of P that same summand is a nonzero local system shifted by $[3](-1)$ and therefore contributes in degree -2 , contrary to (6.3). Supports of dimension at most one do not meet a generic point of U^+ , and a different irreducible divisor cannot contain U^+ .

The local system $\mathcal{V}$ has rank one because its contribution to (6.2) has rank one. Replace U^+ by $U^+ \cap X_{\mathcal{V}}$. Its restriction there is the constant local system found in Lemma 6.1. The inclusion $U^+ \hookrightarrow X_{\mathcal{V}}$ induces a surjection on fundamental groups because both are smooth irreducible complex varieties and U^+ is a nonempty Zariski-open subset. The monodromy of $\mathcal{V}$ is therefore trivial, so the summand is IC_X itself. $\square$

Proposition 6.4 (Alternating support detector). *In $\mathrm{Perv}(P, \mathbf{Q})$, $K = \mathrm{IC}_X$ is a direct summand of ${}^p H^0((K * K)_-)$. After tensoring with $\mathbf{C}$, its image in $\overline{\mathrm{Perv}}(P)$ is nonzero and is a direct summand of $\mathrm{Alt}^2(K)$.*

Proof. The first assertion is Lemma 6.3. Apply the quotient functor to $K \otimes_{\mathbf{Q}} \mathbf{C}$. It is perverse t -exact, convolution is t -exact in $\overline{D}(P)$, and all nonzero perverse-cohomological degrees of $K * K$ map to zero. Hence the image of ${}^p H^0((K * K)_-)$ is the anti-invariant summand of the tensor square of the image of K . Because the support dimension is even, this is $\mathrm{Alt}^2(K)$ under the fibre functor. The exact quotient functor preserves the direct summand supplied by Lemma 6.3; that summand is nonzero because $\chi(K) = 7$. $\square$

7. IDENTIFICATION OF THE FULL TANNAKA GROUP

Let

$$G = G(K), \quad V = \omega(K).$$

By (3.7), $\dim V = 7$. The symmetry (3.4), Verdier self-duality, and the definition of the rigid dual give

$$K^\vee = (-1_P)^* D(K) \simeq K.$$

Thus evaluation is a perfect pairing $K * K \rightarrow \delta_0$. The chosen symmetry is centred at the origin, so no translated skyscraper occurs, and $\dim X = 2$ makes the pairing symmetric. Therefore the full group embeds faithfully as

$$(7.1) \quad G \hookrightarrow O(V).$$

The simple perverse sheaf K is nondivisible: a translation fixing it must fix its support, and (3.8) then makes the translation zero. By [11, Proposition 3.3],

$$(7.2) \quad V|_{G^0} \text{ is irreducible.}$$

This result does not require the ambient abelian variety to be simple.

Choose $\alpha \in \Omega$ as in Proposition 3.6 and use the associated generic-twist fibre functor on the tensor category generated by K . Since any two fibre functors over $\mathbf{C}$ are noncanonically isomorphic, this fibre functor may be identified with ω . Let $G_P = G$, and let G_H be the Tannaka group of the pure Hodge module IC_X^H , acting faithfully on the same complex vector space V . By [14, Corollary 3.2], the Hodge decomposition is defined by a cocharacter $h : \mathbf{G}_m^2 \rightarrow G_H$. The comparison theorem [14, Corollary 2.5] gives

$$G_P \triangleleft G_H, \quad [G_P^0, G_P^0] = [G_H^0, G_H^0] =: G^*.$$

Since $G_P^0 \subset G_H^0$, equation (7.2) implies that $V|_{G_H^0}$ is irreducible. Thus the connected centre Z_H^0 acts by scalar matrices. As G_H^0 is reductive, multiplication $Z_H^0 \times G^* \rightarrow G_H^0$ is surjective with finite kernel.

Put $\mu(z) = h(z, z^{-1})$. This is the antidiagonal reduction used in [14, Section 4.1]; the short argument is recalled here because the passage from the Hodge group to the perverse derived group matters below. Since $\mathbf{G}_m$ is connected, $\mu(\mathbf{G}_m) \subset G_H^0$. By Proposition 3.6,

$$\det \mu(z) = z^{(-2) \cdot 2 + 0 \cdot 3 + 2 \cdot 2} = 1.$$

The determinant is trivial on the semisimple group G^* and therefore descends to a character

$$\overline{\det} : G_H^0 / G^* \rightarrow \mathbf{G}_m.$$

The quotient G_H^0 / G^* is the image of Z_H^0 . On Z_H^0 the determinant is $c \mapsto c^7$; since the faithful scalar action embeds Z_H^0 into $\mathbf{G}_m$, the induced character $\overline{\det}$ has finite kernel. Hence the image of the connected group $\mathbf{G}_m$ under μ in G_H^0 / G^* is both connected and finite, and is therefore trivial. Consequently

$$\mu(\mathbf{G}_m) \subset G^*,$$

the common Hodge and perverse derived group. Its weights on V are

$$(7.3) \quad -2, 0, 2 \quad \text{with multiplicities } 2, 3, 2.$$

Applying the fibre functor to Proposition 6.4 gives

$$(7.4) \quad \mathrm{Hom}_G(V, \Lambda^2 V) \neq 0.$$

Proposition 7.1 (Connected Tannaka group). *There is an isomorphism of pairs*

$$(G^0, V|_{G^0}) \simeq (G_2, V_7).$$

Proof. By (7.1) and (7.2), the connected centre of G^0 acts by orthogonal scalars and is therefore trivial. Thus G^0 is semisimple. An irreducible representation of a product of semisimple groups is an external tensor product. Since 7 is prime and the action is faithful, G^0 has one simple factor.

The classification of irreducible self-dual seven-dimensional representations leaves exactly

SO_7 on its standard module, G_2 on its standard module,

PGL_2 on $\mathrm{Sym}^6(\mathbf{C}^2)$.

Passing to the simply connected cover and consulting the low-dimensional highest weights gives, before imposing self-duality, precisely

$$A_1 : 6\omega_1, \quad A_6 : \omega_1, \omega_6, \quad B_3 : \omega_1, \quad G_2 : \omega_1$$

[5, 7]. The two A_6 modules are dual to each other and are not self-dual; the remaining simple types have no irreducible module of dimension seven. Faithfulness then gives the three group forms displayed above. For SO_7 , the exterior square of the standard module is the irreducible adjoint module, so $\text{Hom}_{SO_7}(V, \Lambda^2 V) = 0$, contrary to (7.4). For a nontrivial cocharacter of PGL_2 , the weights on $\text{Sym}^6(\mathbf{C}^2)$ are seven distinct numbers. For a primitive cocharacter they are

$$-3, -2, -1, 0, 1, 2, 3,$$

and every other nonzero cocharacter multiplies them by a nonzero integer. The trivial cocharacter has one weight of multiplicity seven. Neither possibility agrees with (7.3). Hence $G^0 = G_2$. $\square$

Corollary 7.2 (The full Tannaka group). *There is an isomorphism of pairs*

$$(G, V) \simeq (G_2, V_7).$$

In particular, G is connected; this identifies the full Tannaka group, not only its identity component.

Proof. By Proposition 7.1, $G^0 \simeq G_2$ on V_7 . Since $G^0 \triangleleft G \subset O(V)$,

$$G \subset N_{O(V)}(G_2) = G_2 \times \{\pm I\}.$$

Indeed, conjugation gives a homomorphism from the normalizer to $\text{Aut}(G_2)$. Since $\text{Out}(G_2) = 1$, multiplication by a suitable element of G_2 puts every normalizing element in the centralizer. Schur's lemma makes that centralizer scalar, and its intersection with $O(V)$ is $\{\pm I\}$. If $G \neq G^0$, choose $g \in G \cap (-G_2)$ and write $g = (-I)h$ with $h \in G_2 = G^0$. Then $-I = gh^{-1} \in G$. But $-I$ acts by -1 on V and by $+1$ on $\Lambda^2 V$, contradicting (7.4). Hence $G = G^0 = G_2$. $\square$

Proof of Theorem 1.1. Let $\mathcal{B}^\circ$ be the nonempty open subset obtained in Theorem 5.1. The relative divisor $\mathcal{X} \subset \mathcal{P}$ is symmetric, relatively ample, effective Cartier, flat, and has geometrically integral fibres by the construction at the beginning of Section 4.

Fix $b \in \mathcal{B}^\circ(\mathbf{C})$. The polarization type and the geometric properties of (P_b, X_b) follow from Proposition 3.2 and the construction of L . The locus Ω_b in Proposition 3.6 is nonempty; for any $\alpha_b \in \Omega_b$, that proposition gives Hodge multiplicities $(2, 3, 2)$. The two pair components furnished by Theorem 5.1 imply, through Proposition 6.4, that IC_{X_b} is a direct summand of $\text{Alt}^2(\text{IC}_{X_b})$. Finally, Corollary 7.2 identifies the full Tannaka group and tautological module with (G_2, V_7) . Since b was arbitrary in $\mathcal{B}^\circ(\mathbf{C})$, the theorem follows. $\square$

APPENDIX A. EXACT SHIFTED NORM CERTIFICATE

The polynomial calculations used in Section 4 are contained in the single script `code/g2_shifted_norm_certificate.py`. It starts from the shifted equation and the three sections s_0, s_1, s_2 ; the conic coefficients are solved from the cyclic norm rather than supplied as input. The later formulas for the ordering identity are explicit certificate data: the script substitutes them and verifies the resulting rational-function identities exactly. All calculations use exact SymPy arithmetic.

On the normalized chart $e_1 = v = 1$, write

$$F = z^6 + c_5 z^5 + c_4 z^4 + c_3 z^3 + c_2 z^2 + c_1 z + c_0, \quad k = 3\Delta.$$

Let ℓ be the leading z -coefficient of $F_8(u, 1, z)$ and put $L_c = 4\lambda^2 \ell$. The following explicit formulas form the certificate:

$$\begin{aligned} L_c = & e_2^6 \lambda^2 (u^2 + u + 1) - 3e_2^4 e_3 \lambda^2 (u - 1)(2u + 1) \\ & + 3e_2^3 e_3 \lambda (1 - \lambda)(u + 1) + 9e_2^2 e_3^2 \lambda^2 (u - 1)^2 \\ & + 9e_2 e_3^2 \lambda (\lambda - 1)(u - 1) + 3e_3^2 (\lambda - 1)^2 \\ & + 4c(c + e_2^3 \lambda)(u^2 + u + 1) - 6ce_2 e_3 \lambda (u - 1)(2u + 1) \\ & + 6ce_3 (1 - \lambda)(u + 1). \end{aligned}$$

Define

$$\begin{aligned} N_2^{(0)} &= e_2 \lambda (2e_2^3 \lambda u + e_2^3 \lambda - 6e_2 e_3 \lambda u + 6e_2 e_3 \lambda - 3e_3 \lambda + 3e_3), \\ N_1^{(0)} &= -\lambda (e_2^3 \lambda u + 2e_2^3 \lambda + 3e_2^2 e_3 \lambda u^2 - 6e_2^2 e_3 \lambda u + 3e_2^2 e_3 \lambda \\ &\quad - 3e_2 e_3 \lambda u + 3e_2 e_3 \lambda - 3e_3 \lambda + 3e_3), \\ N_0^{(0)} &= \lambda(u - 1)(-e_2^3 \lambda + 3e_2 e_3 \lambda u - 3e_2 e_3 \lambda + 2e_3 \lambda - 2e_3), \end{aligned}$$

and

$$\begin{aligned} N_2 &= N_2^{(0)} + 2ce_2 \lambda (2u + 1), \\ N_1 &= N_1^{(0)} - 2c\lambda(u + 2), \\ N_0 &= N_0^{(0)} - 2c\lambda(u - 1), \\ b_i &= \frac{N_i}{2L_c} \quad (i = 0, 1, 2). \end{aligned}$$

Set

$$A_c = z^3 + a_2 z^2 + a_1 z + a_0, \quad B_c = b_2 z^2 + b_1 z + b_0,$$

with

$$a_2 = \frac{c_5}{2}, \quad a_1 = \frac{c_4 - a_2^2 - kb_2^2}{2}, \quad a_0 = \frac{c_3}{2} - a_2 a_1 - kb_2 b_1.$$

The remaining three coefficient identities reduce exactly to zero, giving

$$F = A_c^2 + 3\Delta B_c^2.$$

This is the identity used in [Proposition 4.5](#). The same run checks, on the normalized chart,

$$\det(M_s) = (u - v)F_8, \quad \deg F_8 = 8,$$

that F_8 is homogeneous and that $u - v$ occurs with multiplicity one. The scaling in [Lemma 4.1](#) then gives the corresponding identity over K_0 .

For [Lemma 4.3](#), the script reduces

$$(c, \lambda, e_1, e_2, e_3) = (1, 1, 1, 0, 2)$$

modulo 13. It verifies the elliptic and branch discriminants 8 and 4, Rabin's criterion for

$$v^8 - 3v^7 - 2v^6 + 3v^5 + 4v^4 + 4v^3 - 3v^2 - 5v - 2,$$

and the smooth point $(u, v) = (1, 6)$ with gradient $(2, 0)$. For [Lemma 4.10](#), it verifies the section $(u : v : z) = (3 : 5 : 1)$, the conic factorization

$$5 + x + 7Y + 7x^2 + 2Y^2 = 5(1 + 9x + Y)(1 + 12x + 3Y),$$

the fixed-divisor discriminant 1, the three line-section discriminants 4, 3, 5, and the pairwise line intersections

$$(12, 8), \quad (9, 9), \quad (6, 6),$$

all in $\mathbf{F}_{13}$. These finite-field calculations serve only as witnesses for nonempty open conditions; the passage to characteristic zero is the geometric argument in the main text.

The certificate is run by

`python code/g2_shifted_norm_certificate.py`

and its reference output is stored in `code/g2_shifted_norm_certificate.out`.

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